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fMRI and Neuroimaging: Semester Documentation

*Comprehensive Report on fMRI Physics, Preprocessing, Data
Architecture, and Quality Control*

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1. EXECUTIVE SUMMARY

This document outlines the structural organization and metadata contents of the OpenNeuro ds000102 dataset, which utilizes the event-related Flanker task. Before initiating any preprocessing steps in FSL, it is mandatory to understand the Brain Imaging Data Structure (BIDS) format utilized here. This report examines the non-imaging files (.json and .tsv) that provide the critical scanning parameters and behavioral timings required to construct accurate statistical models of the BOLD signal.

2. EXPERIMENTAL DESIGN: THE FLANKER TASK

The OpenNeuro ds000102 dataset employs a widely used cognitive paradigm known as the Eriksen Flanker Task. This task is fundamentally designed to assess cognitive control, conflict resolution, and response inhibition.

During the experiment, subjects are presented with a central directional arrow (the target) flanked by distracting arrows (the flankers) on either side. The subject's objective is to indicate the direction of the central arrow (left or right) via a button press as quickly and accurately as possible, while actively ignoring the flanking arrows.

The task consists of two primary trial conditions:

- **Congruent Trials:** The flanking arrows point in the same direction as the central target (e.g., < < < < or > > > >). This condition presents no cognitive conflict and generally results in faster response times.
- **Incongruent Trials:** The flanking arrows point in the opposite direction of the central target (e.g., < < > < < or > > < > >). This introduces a cognitive conflict, as the brain must actively suppress the automatic visual response to the distractors to focus solely on the target.

The subject is shown a fixation cross in order to focus on the center of the screen, and then either a Congruent or Incongruent Flanker trial is presented for 2000ms. During the trial the subject presses either the left or right button. A jittered interval follows which lasts anywhere from 8,000ms to 14,000ms. (Note that jittered intervals typically increment in seconds; in this case, the jitter for a given trial would be a random selection of one of the following: 8,000ms, 9,000ms, 10,000ms, 11,000ms, 12,000ms, 13,000ms, and 14,000ms) Another fixation cross is presented to begin the next trial.

Behaviorally, incongruent trials consistently produce longer reaction times and lower accuracy rates compared to congruent trials. In the context of this fMRI study, this event-related design allows us to contrast the BOLD signal during incongruent versus congruent trials, effectively isolating the specific neural networks (such as the anterior cingulate cortex) responsible for executive cognitive control.

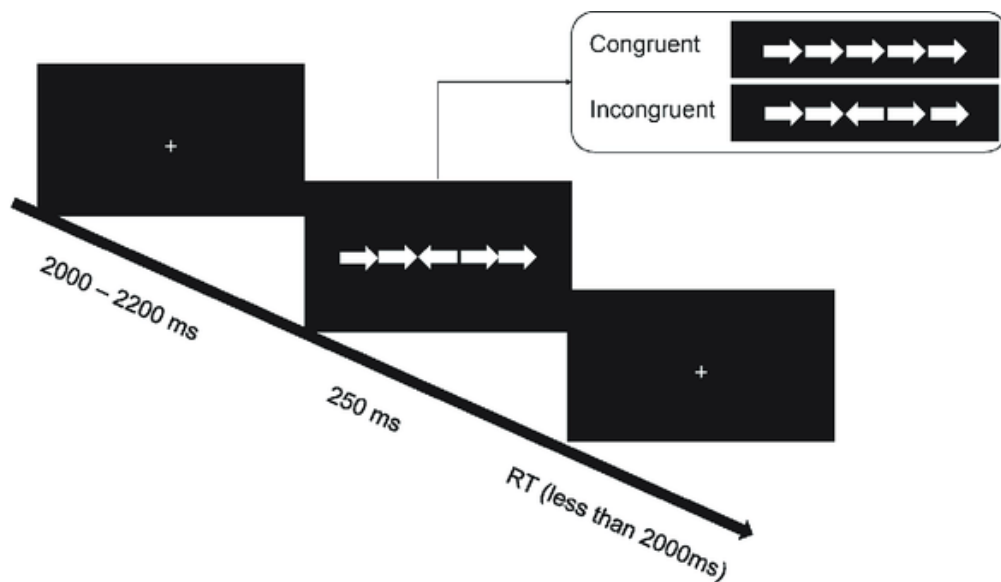


Figure 1: illustration of congruent vs incongruent

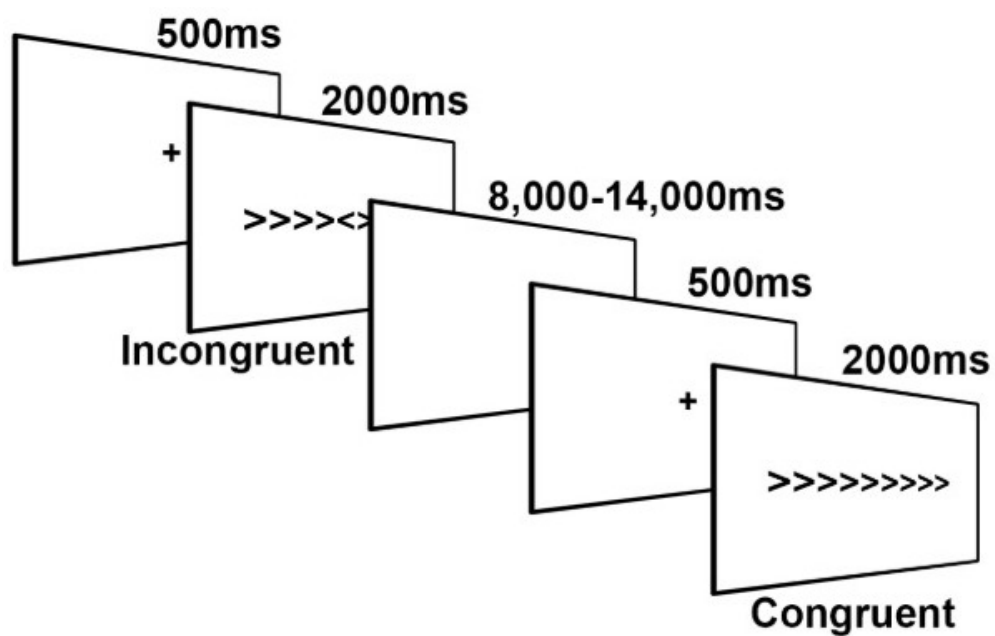


Figure 2: study case of congruent vs incongruent

3. GLOBAL STUDY PARAMETERS (ROOT DIRECTORY)

The root directory houses the overarching metadata that applies uniformly to the entire dataset, ensuring transparency and reproducibility.

- `dataset_description.json`: This file serves as the dataset's fingerprint. It confirms the study name, specifies the BIDS version, and details the licensing. Most importantly, it credits the original researchers (Kelly AMC, et al.), which is essential for academic referencing.
- `participants.tsv`: This tabular file acts as the primary demographic ledger. It maps each `participant_id` (e.g., `sub-01`) to subject-specific variables, notably age and biological sex for the 26 subjects we have.

4. HARDWARE AND ACQUISITION SPECIFICATIONS

To correctly process the raw data in FSL, the software requires exact details about how the MRI scanner captured the images. This data is stored in the root JSON configuration files.

- `task-flanker_bold.json`: This file dictates the physics and timing of the functional T2* runs. The most critical value found here is the **Repetition Time (TR)**, which is registered at 2.0 seconds. The TR dictates the sampling rate of our data and is a mandatory input when setting up the General Linear Model (GLM) in FSL to account for the Hemodynamic Response Function (HRF). It also notes the manufacturer (Siemens Allegra), the Echo Time (TE) of 0.03 seconds, the flipping angle of 80.0 degrees, Magnetic Field Strength of 3.0 T, the Scanning Sequence is Echo Planar and `MRAcquisitionType` which is 2D.
- `T1w.json`: This json file contains the parameters for the high-resolution T1-weighted anatomical scans, acquired using a 3D MPRAGE sequence. Key parameters include a TR of 2.5 seconds, a very short TE of 0.00393 seconds, an Inversion Time of 0.90 seconds, and a Flip Angle of 8.0 degrees. These structural images are required later in the pipeline to register the low-resolution functional images to a standard brain space.

5. RUN-SPECIFIC BEHAVIORAL LOGS (`func` DIRECTORY)

Within individual subject folders (e.g., `sub-01/func`), the `.tsv` files bridge the gap between cognitive psychology and neuroimaging.

5.1. `_events.tsv` Files

These logs are the most critical metadata for task-based fMRI analysis. They record the precise timeline of the experiment inside the scanner and provide the behavioral variables needed for statistical modeling.

- **onset & duration**: These columns map the exact temporal sequence of the run. The `onset` records the exact time (in seconds) a stimulus appeared relative to the start of the scan, explicitly showing the jittered inter-trial intervals. The `duration` confirms that every stimulus was displayed for exactly 2.0 seconds.

- **trial_type:** Rather than just listing the task condition, this dataset utilizes a compound label that merges the stimulus type with the subject's behavioral accuracy for that specific trial (e.g., *congruent_correct* or *incongruent_incorrect*).
- **response_time & correctness:** The `response_time` tracks the subject's physical reaction speed in seconds, a metric that typically increases during the more cognitively demanding incongruent trials. The `correctness` column provides a clear text indicator (*correct* or *incorrect*) of their performance.
- **Stimulus & cond:** These supplementary metadata columns strictly isolate the experimental condition (*congruent* vs. *incongruent*) and assign a corresponding specific statistical condition code (e.g., *cond001* or *cond003*).

6. DERIVATIVES DIRECTORY (QUALITY ASSURANCE)

In BIDS, the `derivatives` folder stores outputs generated from the raw data. For this dataset, it contains pre-computed visual Quality Assurance (QA) PDF reports.

These reports help quickly identify bad data before starting FSL preprocessing:

- **Anatomical QA** (`anatomical_sub-03.pdf`): Displays multi-planar slices of T1-weighted images to verify automated brain extraction (skull-stripping) and segmentation accuracy.
- **Functional QA** (`functional_sub-04.pdf`): Provides a visual summary of the 4D BOLD data, detailing the mean functional signal, spatial standard deviation, and temporal signal-to-noise ratio (tSNR).

Reviewing these PDFs allows researchers to efficiently spot motion artifacts or poor skull-stripping without manually inspecting every `.nii.gz` file.

7. QUANTITATIVE QUALITY CONTROL METRICS

Beyond visual inspection, the `derivatives` directory also contains automated, tabular quality control reports (`aMRIQC.csv` for anatomical and `fMRIQC.csv` for functional scans). These files provide mathematical summaries of image quality.

7.1. Anatomical MRIQC Metrics (`aMRIQC.csv`)

These metrics evaluate the clarity and physical properties of the high-resolution T1-weighted structural scans:

- **SNR (Signal-to-Noise Ratio):** Compares the strength of the actual brain tissue signal against the background scanner noise. *Intuition:* A higher SNR indicates a clearer, less grainy image.
- **CNR (Contrast-to-Noise Ratio):** Measures the distinction between different tissue types (e.g., gray matter vs. white matter). *Intuition:* A higher CNR means sharper boundaries, making automated steps like brain extraction much more accurate.
- **FWHM (Full-Width Half Maximum):** Represents the spatial smoothness or blurriness of the brain image. *Intuition:* A lower FWHM value indicates a crisper, higher-resolution scan.

- **EFC (Entropy Focus Criterion):** A metric used to detect "ghosting" artifacts (faint, overlapping copies of the brain) caused by the subject moving their head during the structural scan. *Intuition:* A lower EFC means less ghosting and motion blur.

7.2. Functional MRIQC Metrics (fMRIQC.csv)

These metrics evaluate the stability of the 4D BOLD signal over time, which is highly susceptible to subject movement:

- **mean_fd (Framewise Displacement):** Measures the average physical distance (in millimeters) the subject's head moved from one 2-second volume to the very next. *Intuition:* High FD values indicate a fidgety subject. Since we are looking for tiny BOLD changes, excessive head motion can easily ruin the task signal.
- **dvars (Derivative of Variance):** Measures how much the actual voxel intensity changes from one frame to the next. *Intuition:* If the whole brain's signal suddenly spikes or drops, DVARS shoots up. This metric perfectly correlates with sudden head jerks or scanner artifacts.
- **m_tsnr (Mean Temporal Signal-to-Noise Ratio):** Evaluates how stable the BOLD signal remains over the entire 5-minute task block. *Intuition:* A higher tSNR means the scanner recorded a clean, stable baseline signal, making it much easier to detect the subtle neurological responses caused by the Flanker task.
- **outlier & perc_fd:** Represent the fraction of timepoints (volumes) that are corrupted by excessive motion spikes. *Intuition:* A high percentage means a significant portion of the run is unusable and may need to be scrubbed.

8. BIDS STANDARD AND DATA ORGANIZATION

As indicated by the successful validation badge on the OpenNeuro dataset page, the ds000102 dataset strictly adheres to the Brain Imaging Data Structure (BIDS).

8.1. What is BIDS and Why is it Used?

Historically, neuroimaging labs organized their data using ad-hoc, lab-specific naming conventions and directory structures. This lack of standardization made sharing data across institutions incredibly difficult and hindered reproducibility. BIDS was developed as a simple, intuitive, and highly structured standard to resolve this. By mandating exactly how folders must be nested, how files must be named, and what metadata must be included, BIDS enables researchers to easily understand new datasets. More importantly, it allows developers to build automated, universal analysis pipelines (such as MRIQC or fMRIPrep) that can automatically parse the data without requiring manual user input or reconfiguration.

8.2. Relation to Dataset ds000102

- **Standardized Directory Hierarchy:** BIDS mandates that data is separated by subject and modality. This is why we observed individual `sub-XX` folders containing dedicated `anat` (anatomical) and `func` (functional) subdirectories.

- **Decoupled Metadata (.json):** Instead of hiding critical scanner parameters (like the TR or Echo Time) deep within complex NIfTI or DICOM image headers, BIDS requires them to be easily accessible in human-readable JSON sidecars (e.g., `task-flanker_bold.json` and `T1w.json`). This allows scripts to grab the TR without having to load a 2-gigabyte image file into memory.
- **Tabular Task Timings (.tsv):** The BIDS standard specifies that all behavioral data must be decoupled from the raw images and stored as plain text. The `_events.tsv` files we analyzed in the `func` directory ensure that the onset times, durations, and conditions of the Flanker task are universally readable by any statistical software (like FSL or SPM) to build the General Linear Model.
- **Global Context:** Global files like `dataset_description.json` and `participants.tsv` sit at the root directory, fulfilling the BIDS requirement to provide immediate context, licensing, and demographic control variables for the entire study cohort.

9. PHYSICAL FOUNDATIONS OF FMRI

9.1. T_2^* Relaxation and the BOLD Signal

Functional Magnetic Resonance Imaging (fMRI) primarily relies on the Blood-Oxygen-Level-Dependent (BOLD) contrast. The physical mechanism driving this contrast is T_2^* (T_2 -star) relaxation. While standard T_2 relaxation measures the decay of transverse magnetization due to random spin-spin interactions, T_2^* also accounts for the accelerated dephasing caused by fixed local magnetic field inhomogeneities.

Deoxygenated hemoglobin is highly paramagnetic, which distorts the local magnetic field and speeds up spin dephasing, leading to a faster T_2^* signal decay (a darker image). Conversely, oxygenated hemoglobin is diamagnetic and does not distort the field. When a brain region activates, there is an overcompensation of oxygenated blood delivered to the area. This flushes out the deoxygenated hemoglobin, reducing magnetic field inhomogeneities, slowing down the T_2^* decay, and resulting in a brighter measurable signal on the scanner.

9.2. The Hemodynamic Response (HDR) and Neurovascular Coupling

While the BOLD contrast relies on the magnetic properties of hemoglobin, the biological bridge between neural activity and blood flow is known as **neurovascular coupling**. When neurons fire, they consume local oxygen and release neurotransmitters like glutamate. Astrocytes, a type of glial cell, detect this synaptic activity and release vasodilatory chemical signals to nearby arterioles. This triggers a localized vasodilation, causing a massive influx of oxygenated blood that overcompensates for the initial metabolic demand.

The Hemodynamic Response (HDR)—also known as the Hemodynamic Response Function (HRF)—describes the precise temporal profile of this blood-flow change following a brief neural event. The HDR spans several distinct phases:

- **Initial Dip:** A brief, subtle decrease in the BOLD signal (often occurring within the first 1-2 seconds) caused by a localized spike in oxygen consumption by active neurons before the fresh blood supply arrives.

- **Peak:** The massive influx of oxygenated blood overcompensates for the demand, causing the BOLD signal to rise and peak roughly 4 to 6 seconds after the initial stimulus.
- **Undershoot:** As neural activity ceases, blood flow decreases faster than blood volume, leading to a temporary accumulation of deoxygenated blood. The signal dips below the initial baseline for several seconds.
- **Return to Baseline:** The cerebral blood flow and volume eventually stabilize, returning the signal to resting levels approximately 15 to 30 seconds after the stimulus.

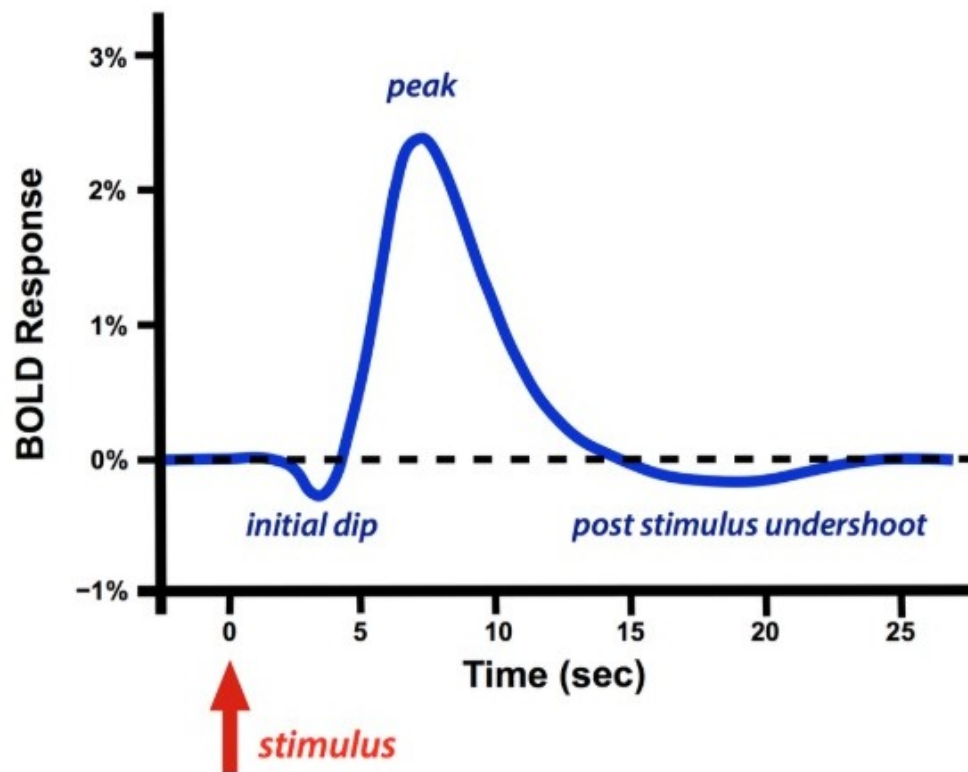


Figure 3: BOLD response illustration

Deriving the Ideal HRF for Statistical Modeling

To analyze fMRI data using a General Linear Model (GLM), we must predict what the BOLD signal *should* look like if a specific brain region is actively responding to our task. We achieve this by defining an **Ideal HRF**.

Empirical studies have shown that the canonical (ideal) HRF can be mathematically approximated using a **double-gamma function**. The first gamma function models the main positive peak of the blood flow response, while the second, smaller gamma function models the delayed post-stimulus undershoot.

In practice, to build our statistical model, we take the exact stimulus onset times and durations from our behavioral logs (the `_events.tsv` files) and **convolve** them with this ideal double-gamma HRF. This mathematical convolution translates discrete, instantaneous cognitive events into a continuous, sluggish hemodynamic prediction. This predicted time-series becomes the primary regressor in our GLM, allowing the software to identify voxels whose actual measured BOLD signal significantly matches our ideal prediction.

Conceptual Exercises (Andy's Brain Book)

To solidify the theoretical understanding of the BOLD signal and its mathematical modeling, the following conceptual exercises from Andy's Brain Book (Chapter 3) were addressed:

Exercise 1: Distinguishing the BOLD Signal and the HRF

While often used interchangeably, these terms represent distinct concepts in fMRI analysis:

- **BOLD Signal:** This is the actual, empirical, and measurable change in the magnetic resonance signal captured by the scanner. It is driven by the physical ratio of oxygenated to deoxygenated blood in the brain over time.
- **Hemodynamic Response Function (HRF):** This is the theoretical, mathematical model (typically a Gamma distribution) used to approximate the ideal shape of the BOLD response to a single, instantaneous neural event. It acts as a basis function for our statistical predictions.

Exercise 2: Simulating Convolution of Different Stimulus Shapes

The shape of the predicted BOLD response changes drastically depending on the duration of the stimulus, as demonstrated through convolution simulations:

- **Impulse (Dirac) Stimulus:** Represents a stimulus with effectively zero duration (e.g., a rapid finger snap). Convolution with the HRF returns the canonical, unadulterated Gamma distribution shape, featuring a sharp peak and a long right tail.
- **Boxcar (Rectangle) Stimulus:** Represents a sustained stimulus (e.g., viewing a flashing checkerboard for 15 seconds). Convolution with the HRF results in a moving average. The predicted response broadens significantly, maintaining a sustained plateau proportional to the stimulus duration before eventually descending back to baseline. Furthermore, if multiple stimuli occur rapidly, their individual HRFs overlap and sum together, creating a complex, continuous waveform.

10. STANDARD FMRI PREPROCESSING PIPELINE

Raw fMRI images are inherently noisy and riddled with artifacts. Before constructing a General Linear Model (GLM), the data must undergo a rigorous sequence of data-cleaning steps known as preprocessing. The standard pipeline includes:

- **Brain Extraction (Skullstripping):** The removal of non-brain tissues (skull, fat, and skin) from the images to ensure that subsequent algorithms only analyze cerebral voxels.
- **Motion Correction:** Correcting for subject head movement during the scan. The software registers every functional volume in the time series to a single reference volume, estimating and correcting translations and rotations across the X, Y, and Z axes.
- **Slice-Timing Correction:** Because the scanner acquires slices of the brain sequentially (not instantly), different slices of the same volume are recorded at slightly different times. This step interpolates the data so that all slices in a single TR appear to have been acquired exactly simultaneously.

- **Smoothing:** Applying a spatial low-pass filter (typically a Gaussian kernel) to blur the images intentionally. This averages out random thermal noise, boosts the Signal-to-Noise Ratio (SNR), and helps compensate for minor anatomical differences between subjects.
- **Registration and Normalization:** Aligning the low-resolution functional images to the subject's high-resolution T1 structural scan (Registration), and then warping that combined image into a standardized brain space, such as the MNI152 template (Normalization). This allows for accurate group-level statistics.

10.1. Case Study: FEAT Pipeline Application on Subject 08

To demonstrate the practical application of these preprocessing steps, the pipeline was executed on real cases (Sub-08, Sub-07, Sub-22, Sub-03) 4 subjects for two runs from the Flanker dataset and i will mention (sub-08) as a study case using FSL's FMRI Expert Analysis Tool (FEAT).

10.1.1 Brain Extraction (BET) Optimization

Brain extraction was performed using FSL's Brain Extraction Tool (BET). The fractional intensity threshold (f) required manual tuning to ensure optimal removal of non-brain tissues without eroding cortical margins. Visual inspection in FSLeaves revealed that the default threshold of $f = 0.5$ resulted in poor skull stripping, incorrectly excluding valid brain tissue. Adjusting the threshold to $f = 0.2$ produced a robust and accurate extraction.



Figure 4: Comparison of BET skull stripping in FSLeaves. Left: Poor extraction with threshold 0.5. Right: Corrected extraction with threshold 0.2.

10.1.2 Motion Correction and Temporal Filtering

Motion correction was applied using MCFLIRT. Subject 08 demonstrated high physical stability during the run. The quantitative assessment reported a mean absolute displacement of 0.08 mm and a mean relative displacement of 0.07 mm. Additional temporal preprocessing included slice-timing correction using Fourier-space time-series phase-shifting and highpass temporal filtering (Gaussian-weighted least-squares straight line fitting) with a sigma of 50.0 seconds.

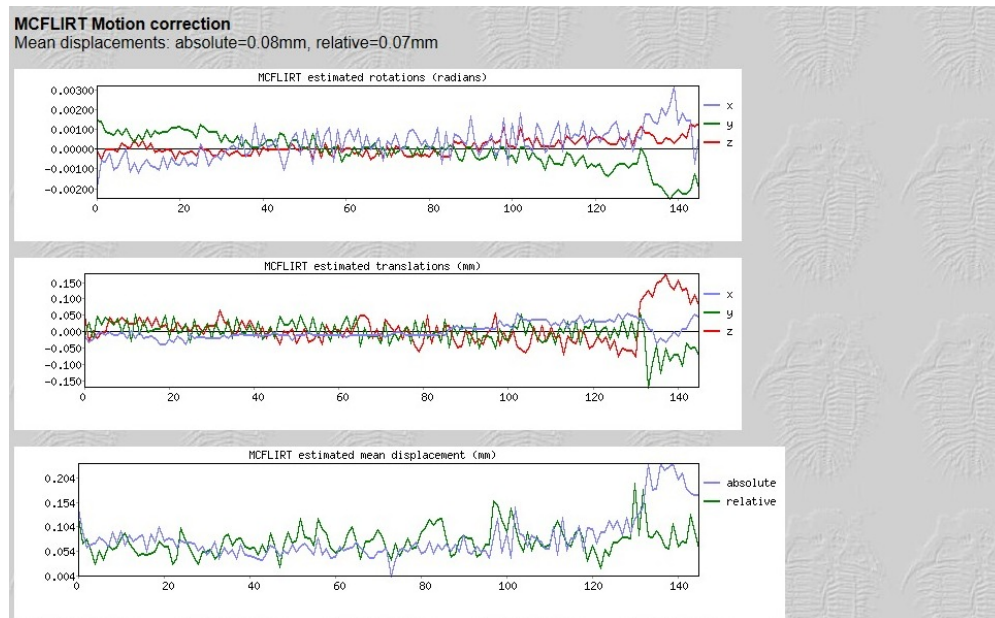


Figure 5: MCFLIRT motion correction estimated rotations, translations, and mean displacement for Subject 08.

10.1.3 Spatial Smoothing

To boost the Signal-to-Noise Ratio (SNR) and average out random thermal noise, spatial smoothing was applied using a Gaussian kernel with a Full-Width Half Maximum (FWHM) of 5 mm. Grand-mean intensity normalization of the entire 4D dataset was also performed by a single multiplicative factor.

10.1.4 Registration and Normalization

Registration of the functional data to high-resolution structural and standard space images was carried out using FLIRT. The FEAT pipeline successfully aligned the low-resolution `example_func` to the high-resolution T1 structural scan, and subsequently normalized the high-resolution structural image to the MNI152 standard space template.

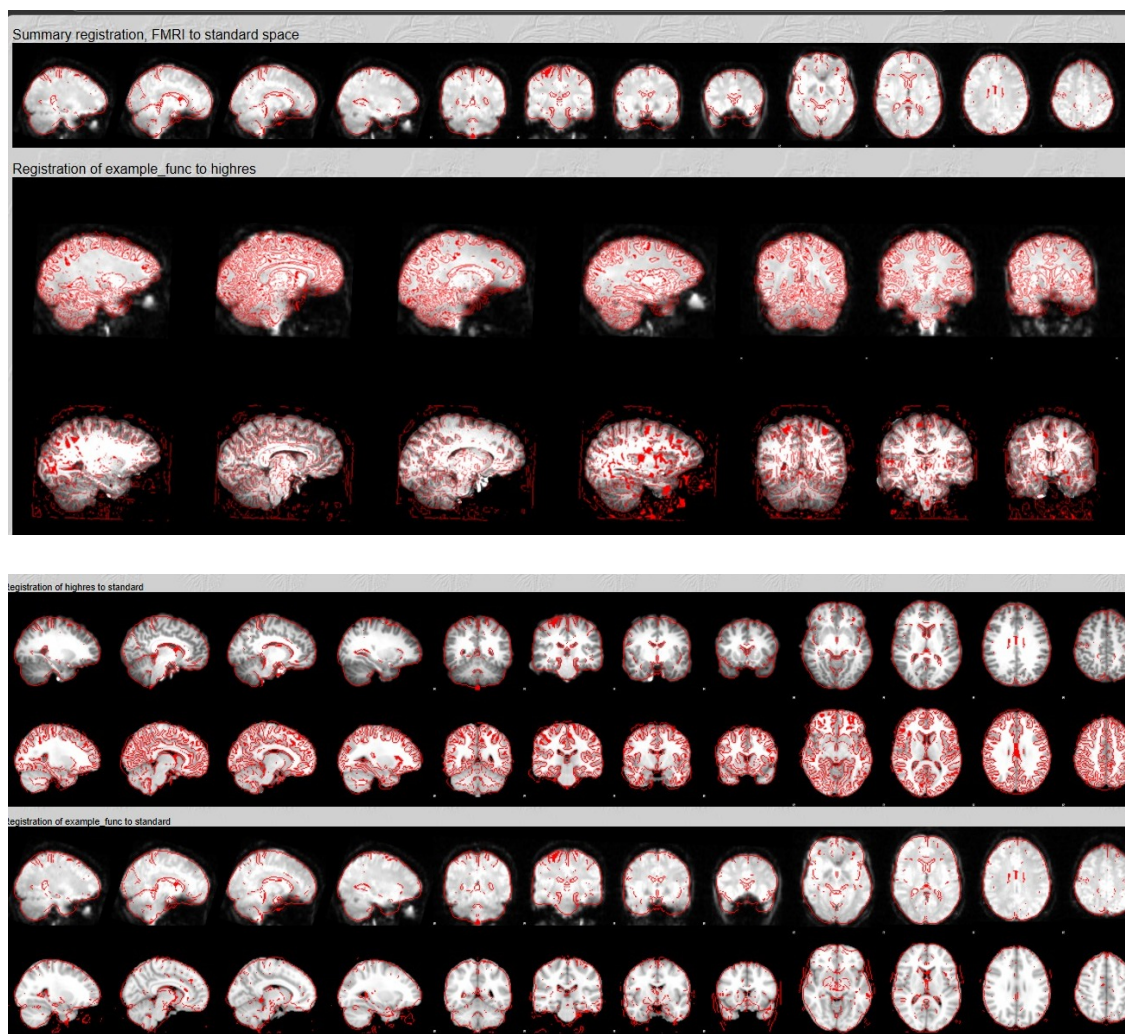


Figure 6: FLIRT registration summary. Top: Registration of `example_func` to `highres`. Bottom: Registration of `highres` to standard MNI space.

10.1.5 Manual Visual Validation of Registration

While the automated FEAT report provides static summary images of the registration, best practices dictate a manual visual inspection utilizing FSLeyes to confirm alignment accuracy. This is achieved by overlaying the warped output files onto their respective target templates from the `reg` directory.

- Functional to Anatomical Validation:** The subject's structural image (`highres.nii`) was loaded as the underlay, with the registered functional image (`example_func2highres.nii`) loaded as a semi-transparent overlay. Inspection confirmed that the internal structures, particularly the lateral ventricles, and the outer cortical boundaries were tightly aligned, indicating a successful rigid-body registration.
- Functional to Standard Space Validation:** The MNI152 standard template (`standard.nii`) was loaded as the underlay, with the fully normalized functional image (`example_func2standard.nii`) as the overlay. The functional data accurately matched the global geometry of the MNI template, confirming that the non-linear warping (or 12 DOF affine transformation) successfully mapped the subject's unique neuroanatomy into standardized space.

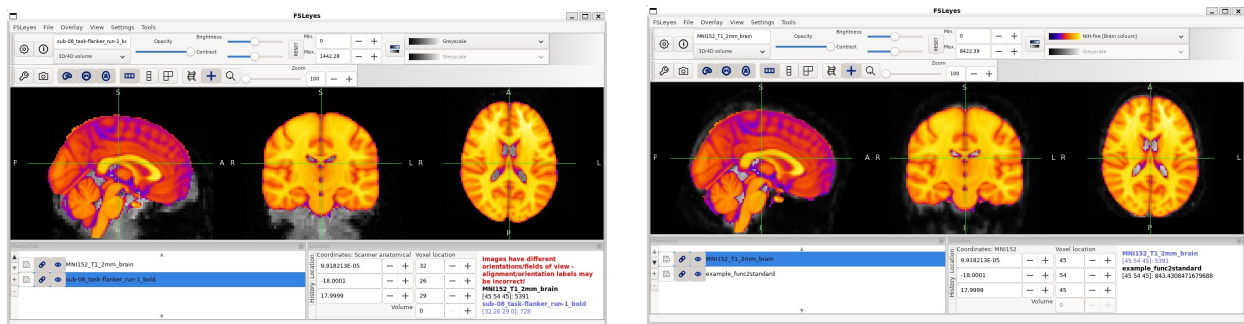


Figure 7: Manual registration validation in FSLeyes. Left: `sub-08_task-flanker_run-1_bold.nii` raw data (underlay) to standard MNI template. Right: `example_func2standard` (underlay) successfully warped to the standard MNI template.

10.2. Parameter Exploration and Practical Exercises (Andy's Brain Book)

To thoroughly understand the sensitivity of the preprocessing algorithms, several parameter exploration exercises were conducted on Subject 08 as outlined in Andy's Brain Book (Tutorial 4).

Exercise: Extreme BET Thresholds on Functional and Anatomical Data

The Brain Extraction Tool (BET) relies heavily on the fractional intensity threshold (f). We applied extreme thresholds ($f = 0.1$ and $f = 0.9$) to both the functional data and the high-resolution anatomical T1-weighted scan (sub-08_T1w.nii.gz).

- **Threshold 0.1 (Lenient):** At $f = 0.1$, the algorithm is highly conservative about removing voxels. For both functional and anatomical scans, this resulted in a vast amount of non-brain tissue (skull, dura mater, and neck muscles) being left in the image.
- **Threshold 0.9 (Aggressive):** At $f = 0.9$, the algorithm violently strips away voxels. This resulted in the catastrophic loss of valid brain tissue, effectively cropping out large portions of the cerebral cortex.

Conclusion: Neither extreme is suitable. However, if forced to choose one, the $f = 0.1$ image is preferable. Leaving non-brain tissue can potentially be masked out later in the pipeline, whereas data deleted by $f = 0.9$ is permanently lost. Through this exploration, we confirmed that a threshold of $f = 0.2$ to 0.3 serves as the optimal "sweet spot" for this specific dataset.

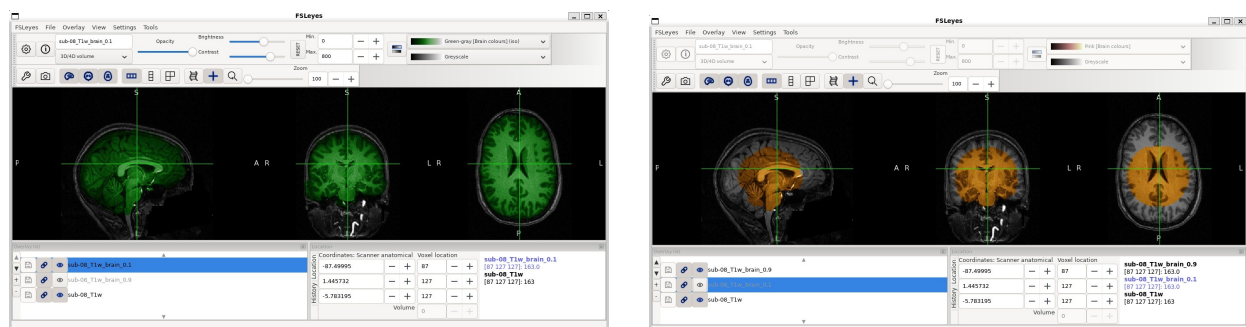


Figure 8: Exercise 3: BET on Subject 08 Anatomical T1w. Left: Lenient threshold ($f = 0.1$) leaves skull and neck. Right: Aggressive threshold ($f = 0.9$) deletes valid cortical tissue.

Exercise: Visual Inspection utilizing FSLeaves Montages

To accurately verify the success of the skull stripping, the overlay contrast colors were manipulated in FSLeaves (e.g., switching from standard grayscale to 'Red-Yellow'). Utilizing the zoom slider allows for a targeted inspection of the boundaries, specifically at the frontal and parietal margins where the dura mater is notoriously difficult for BET to separate from the cortex.

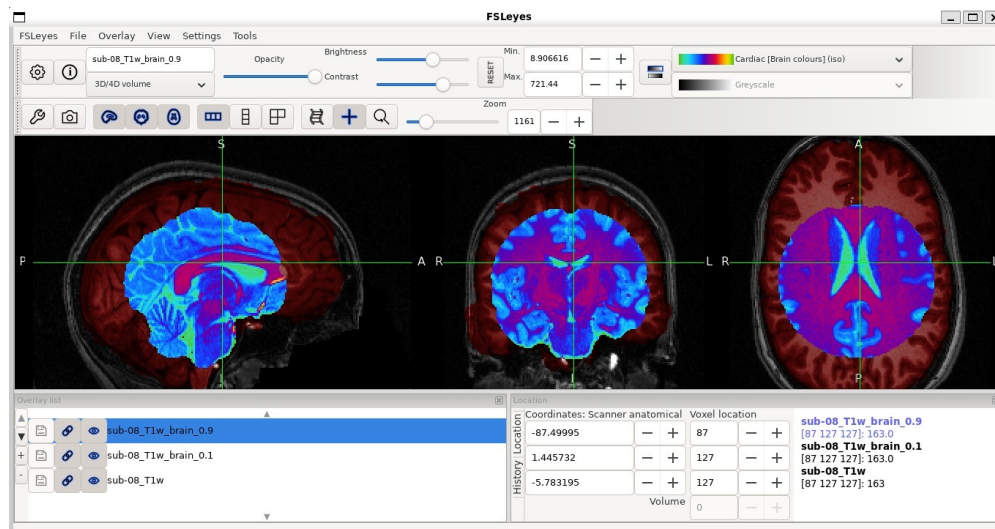


Figure 9: Exercise 2: FSLeaves view utilizing high-contrast overlay colors and localized zooming to verify cortical boundaries.

Exercise: Spatial Smoothing Kernel Variations (3mm vs. 12mm)

Spatial smoothing intentionally blurs the image to boost the Signal-to-Noise Ratio (SNR). We tested two alternative Full-Width Half Maximum (FWHM) kernels on Run 1:

- **3mm Kernel:** Expectedly, this preserved high spatial resolution but failed to adequately average out thermal noise, leaving the image looking grainy.
- **12mm Kernel:** This applied a massive blur. While SNR is technically higher, the functional data was washed out, destroying spatial localization. A functional activation in this image would bleed heavily into adjacent, non-active anatomical regions.

Comparing these extremes in FSLeaves confirms our prediction: a 5mm kernel (used in the main pipeline) is the ideal compromise between noise reduction and spatial fidelity.

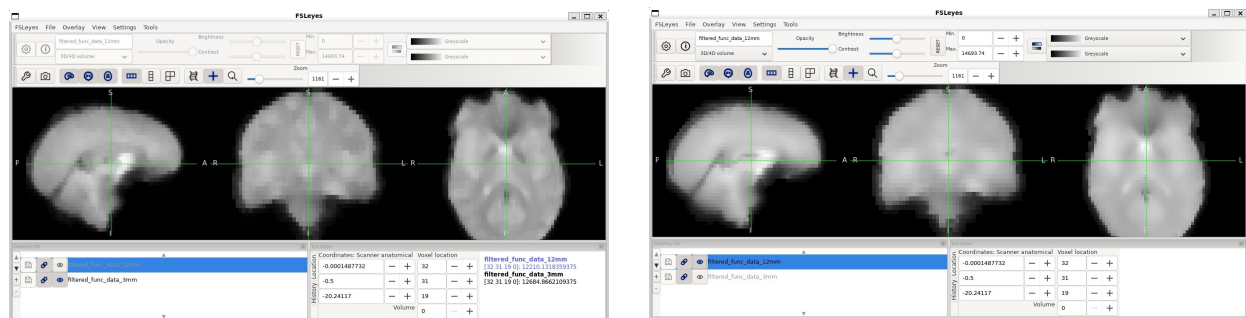


Figure 10: Exercise 5: Comparison of filtered_func_data.nii.gz. Left: 3mm smoothing kernel (grainy). Right: 12mm smoothing kernel (heavily blurred and washed out).

Exercise: Registration Degrees of Freedom (3 DOF vs. 12 DOF)

When registering Run 1 using only 3 Degrees of Freedom (DOF), the output was drastically misaligned compared to 12 DOF. **Why this occurs:** 3 DOF strictly limits the transformation to rigid *translations* (moving X, Y, Z). It cannot rotate, scale, or shear the image. Because human brains vary in size and shape, warping a subject's functional image into the standardized MNI152 template fundamentally requires scaling and shearing. Standard registration utilizes 12 DOF to permit these complex geometric transformations, ensuring accurate anatomical alignment across the cohort.

Exercise: Boundary-Based Registration (BBR) vs. 12 DOF

Standard 12 DOF FLIRT registration aligns functional and structural images by comparing whole-image intensity cost functions (like the correlation ratio). We reran the registration utilizing Boundary-Based Registration (BBR). **The Difference:** BBR is a highly advanced algorithm that specifically aligns images by maximizing the intensity gradients across distinct tissue boundaries—most notably the boundary between gray matter and white matter. **The Case for BBR:** When justifying a pipeline, BBR is vastly superior for EPI-to-T1 (functional-to-structural) registration. Because functional images suffer from spatial distortions and signal dropouts, relying on anatomical boundaries (BBR) rather than overall brightness (12 DOF) results in a much tighter, highly accurate alignment.

11. QUALITY CONTROL (QC) ANALYSIS

Automated QC metrics evaluate data integrity, allowing researchers to exclude corrupted time-points or apply denoising regressors.

11.1. Purpose of Quality Control

fMRI data is heavily impacted by scanner drift and head motion. QC ensures that observed BOLD signal changes are task-related and not driven by these artifacts. Functional QC metrics evaluate temporal stability utilizing Framewise Displacement (FD), the Derivative of Variance (DVARS), and temporal SNR (tSNR).

11.2. Flanker Dataset QC Assessment

Table 1 attached below presents the qualitative visual quality control assessment extracted for all 26 subjects in the Flanker dataset. The visual motion level is categorized into low, medium, and high, accompanied by precise observational notes regarding the volume numbers where artifacts were detected.

Table 1: Visual Quality Control Assessment (Flanker Dataset)

Sub.ID	Run.num	Visual Motion Level	General Notes
sub-17	run-2	low	very slight up/down motion around volume 35/50
sub-07	run-1	low	stable, almost no motion artifact
sub-14	run-2	high	motion artifact, subject's head went up significantly around volume 112
sub-04	run-1	low	stable, almost no motion artifact
sub-11	run-1	medium	the subject's head jittered slightly around volume 74 but recovered quickly
sub-26	run-2	low	subject's head moved around volume 40/111
sub-3	run-1	medium	heavy motion back and forth in volume 67
sub-15	run-2	medium	very slight up/down motion around volume 34/100
sub-25	run-1	high	subject's head moved obviously on volume 38/87 and recovered
sub-02	run-2	low	stable, almost no motion artifact
sub-19	run-1	high	subject's head moved so significantly from 1 to 10, and from 42 to 51
sub-18	run-2	low	stable
sub-05	run-1	low	almost no motion artifact
sub-01	run-2	low	stable case
sub-12	run-1	low	stable has a very slight movement on volume 45
sub-23	run-2	low	no motion artifacts
sub-13	run-1	low	stable
sub-16	run-2	medium	slight motion on volumes 70/121
sub-06	run-1	low	slight motion on volume 89
sub-8	run-2	high	motion artifact, subject's head moved a lot in the first 30 volumes
sub-10	run-2	medium	continuous motion at the jaw
sub-09	run-1	low	stable
sub-24	run-1	low	very slight motion on 109
sub-20	run-2	medium	motion artifact around volume 133
sub-22	run-1	low	slight motion around 104
sub-21	run-2	medium	motion artifact around 98

11.3. Qualitative Insights and Artifacts

Analyzing the aggregated metrics from Table 1 provides a macroscopic view of the cohort's data integrity. A review of the dataset reveals several key technical insights:

- **Subject Variability in Motion:** There is observable variance in Framewise Displacement (FD) across the 26 subjects. The average FD across the cohort is 0.106 mm. While some subjects remained remarkably still (e.g., sub-23 recorded an FD of just 0.061 mm), others exhibited notably higher average head motion (e.g., sub-08 with a peak FD of 0.147 mm).
- **Impact of Motion on Signal Quality:** The data demonstrates a clear inverse relationship between subject movement and temporal stability. Subjects who recorded higher average motion and higher variance (like sub-08, with a peak DVARS of 30.80) generally exhibited a much lower Temporal Signal-to-Noise Ratio (tSNR of 55.50). Conversely, stable subjects like sub-20 (lowest DVARS of 18.06) achieved the highest tSNR in the cohort (89.28). This confirms that head motion directly degrades BOLD signal quality.
- **Consistency in Spatial Resolution:** In contrast to the highly variable temporal metrics, the spatial smoothness of the images, represented by the Full-Width Half Maximum (FWHM), remained tightly clustered around an average of 1.85 (ranging only from 1.69 to 2.01). This suggests that the scanner's hardware performance and acquisition sequence were highly stable; the primary source of variance was subject movement.

12. FIRST-LEVEL ANALYSIS: STATISTICAL MODELING (GLM)

12.1. Calculations and Equations Used

To determine which areas of the brain are activated by the Flanker task, we fit a General Linear Model (GLM) to the preprocessed functional data. The GLM attempts to explain the observed BOLD time-series at every single voxel as a linear combination of expected responses (our experimental conditions) plus unexplained noise.

The fundamental equation of the GLM applied here is:

$$Y = \beta_1 X_1 + \beta_2 X_2 + \epsilon \quad (1)$$

- **Y (Dependent Variable):** The actual, observed fMRI time-series data for a specific voxel (measured in arbitrary scanner units over time).
- **X (Design Matrix / Regressors):** The expected "Ideal Time-Series." In our case, X_1 represents the incongruent condition and X_2 represents the congruent condition. These are created by mathematically convolving the exact stimulus onset times (from the `_events.tsv` files) with the ideal Hemodynamic Response Function (HRF).
- **β (Beta Weights / Parameter Estimates):** The scaling factors estimated by the GLM software. The algorithm calculates how much the ideal model (X) needs to be multiplied by to best fit the actual data (Y).
- **ϵ (Residuals):** The error term. This represents all the variance in the BOLD signal that the model could not explain (e.g., scanner drift, unmodeled head motion, or physiological noise).

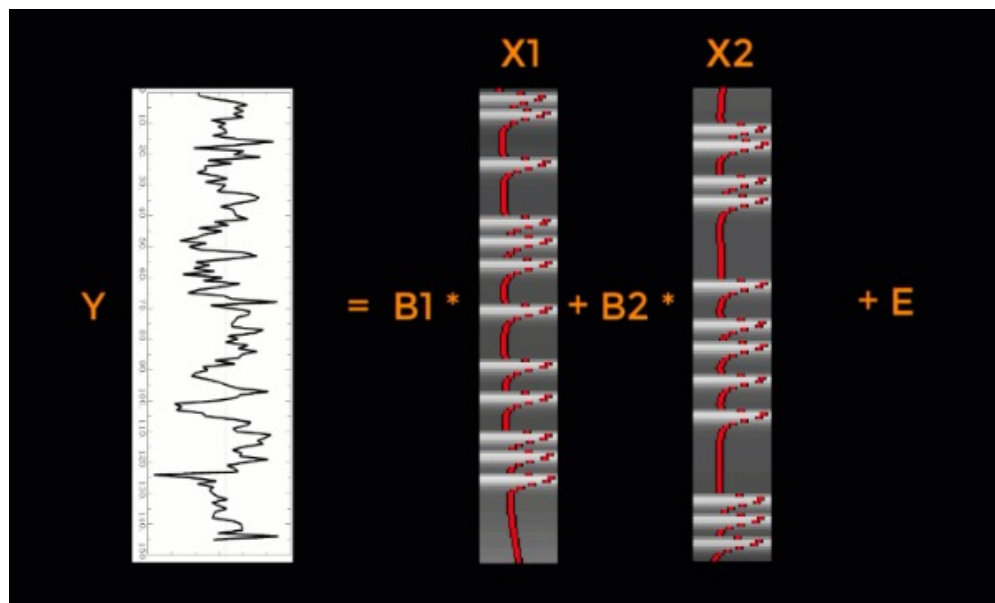


Figure 11: Fitting the time series to fMRI data

12.2. Statistical Inference: The Null Hypothesis and Null Distribution

To determine if the calculated β weights represent genuine neural activation or just random scanner noise, we utilize Null Hypothesis Significance Testing (NHST). In fMRI analysis, the **Null Hypothesis** (H_0) assumes that the experimental stimulus has absolutely no effect on the BOLD signal (i.e., the true β weight is exactly 0).

Calculating the Null Distribution

Before we can determine if an activation is statistically significant, we must mathematically define what "random noise" looks like. This baseline model is called the **Null Distribution**. In fMRI, this distribution is typically calculated in one of two ways:

- **Analytically (Parametric):** Assuming the background noise in the scanner follows a standard Gaussian (bell curve) shape. The software automatically maps the expected variance to a Standard Normal Distribution (where the mean is 0, representing zero task-related brain activation).
- **Empirically (Permutation Testing):** By taking the actual behavioral timing files (the `_events.tsv`) and randomly shuffling or shifting the onset times thousands of times. By running the GLM on these "fake" random timings, the software builds a custom mathematical distribution of what purely coincidental BOLD fluctuations look like for that specific run.

The Coin Flip: Constructing the Null Distribution

To understand how a Null Distribution is mathematically constructed, consider the classic **Coin Flip Experiment**. If you flip a coin 10 times, how do we build the distribution of expected outcomes under the null hypothesis (which states the coin is perfectly fair, $p = 0.5$)?

We obtain this distribution through combinatorial mathematics (specifically, a Binomial Distribution):

1. **Total Possible Outcomes:** Every flip has 2 possible outcomes (Heads or Tails). For 10 independent flips, there are $2^{10} = 1024$ totally unique sequences of heads and tails.
2. **Calculating Individual Probabilities:** We calculate the number of ways to get exactly k heads (where k ranges from 0 to 10) using the combinations formula $\binom{n}{k}$.
 - For example, there are 252 different ways to get exactly 5 heads. The probability is $252/1024 \approx 0.246$ (the peak of the distribution).
 - There is only exactly 1 way to get 10 heads (HHHHHHHHHH). The probability is $1/1024 \approx 0.00097$.
3. **Building the Curve:** By calculating and plotting the probability for every single possible outcome ($k = 0$ through $k = 10$), we construct a symmetrical, bell-shaped graph. **This graph is our Null Distribution.** It maps out exactly what random chance looks like.

Understanding Cumulative Probability (The P-Value): If we actually conduct the experiment and get 10 heads in a row, we look at our constructed null distribution. The p-value is the **cumulative probability** of observing a result *at least as extreme* as the one recorded.

If we observed 9 heads, the p-value would be calculated by adding the probability of getting exactly 9 heads *plus* the probability of getting 10 heads. Because the probability of getting 10 heads is so incredibly low ($p < 0.001$, residing in the far extreme tail of our constructed distribution), we reject the null hypothesis and conclude the coin is rigged.

Calculating the P-Value in fMRI (Area Under the Curve)

In our fMRI General Linear Model, the software calculates continuous statistics rather than discrete coin flips. Once the GLM computes the Z-statistic for a specific voxel (derived from dividing the COPE by the VARCOPE), the exact p-value is mathematically calculated using the **integral (area under the curve)** of our Null Distribution.

- **Mapping the Statistic:** We map the voxel's observed Z-score onto the standard bell curve of our Null Distribution.
- **The Integral (Tail Area):** The exact p-value is calculated by taking the integral of this probability density function, starting from our observed Z-score out to infinity.
- **Applying the Threshold:** For example, if a voxel achieves a Z-score of 3.1, calculating the area under the extreme tail of the curve beyond 3.1 yields a p-value of approximately 0.001. Because this calculated area is smaller than our established alpha threshold ($\alpha = 0.05$), we confidently reject H_0 and declare the voxel significantly active.

12.3. The Relationship Between Beta and P-Value

It is critical to distinguish between β weights and p -values, as they represent entirely different statistical concepts:

- **Beta (β):** Represents the *Effect Size* or signal magnitude. It tells us how large the BOLD response was.
- **P-value (or Z-score):** Represents *Statistical Significance*. It tells us the probability of observing that β weight purely by chance.

A large β weight does not automatically guarantee a significant p -value. The relationship is governed by the variance (noise) in the data, linked by the t -statistic:

$$t = \frac{\beta}{SE(\beta)} \quad (2)$$

Where $SE(\beta)$ is the Standard Error (the noise). If a voxel has a massive BOLD response (β), but the voxel is also incredibly noisy (large SE), the resulting t -statistic will be small, and the p -value will not reach significance.

13. FIRST-LEVEL ANALYSIS IMPLEMENTATION

13.1. Cohort Execution and Required Exercises

First-level analysis was successfully executed for 4 subjects (sub-03, sub-07, sub-08, sub-22) across 2 runs each, resulting in 8 distinct FEAT directories.

13.2. Complete Test Case: Subject 08, Run 1

To detail the FEAT pipeline, sub-08 run-1 is utilized as the primary test case. The fully preprocessed `filtered_func_data.nii.gz` was loaded into the FEAT Statistics GUI.

1. Model Setup (EVs): Two Explanatory Variables (EVs) were established using the Custom (3-column format) timing files created earlier:

- **EV1:** `incongruent_run1.txt`
- **EV2:** `congruent_run1.txt`

2. Contrasts: To extract statistical meaning, three contrasts were defined:

- **C1 (Incongruent):** `[1 0]` - Tests for average activation during incongruent trials.
- **C2 (Congruent):** `[0 1]` - Tests for average activation during congruent trials.
- **C3 (Incongruent - Congruent):** `[1 -1]` - The primary contrast of interest, isolating the specific cognitive effort required for conflict resolution.

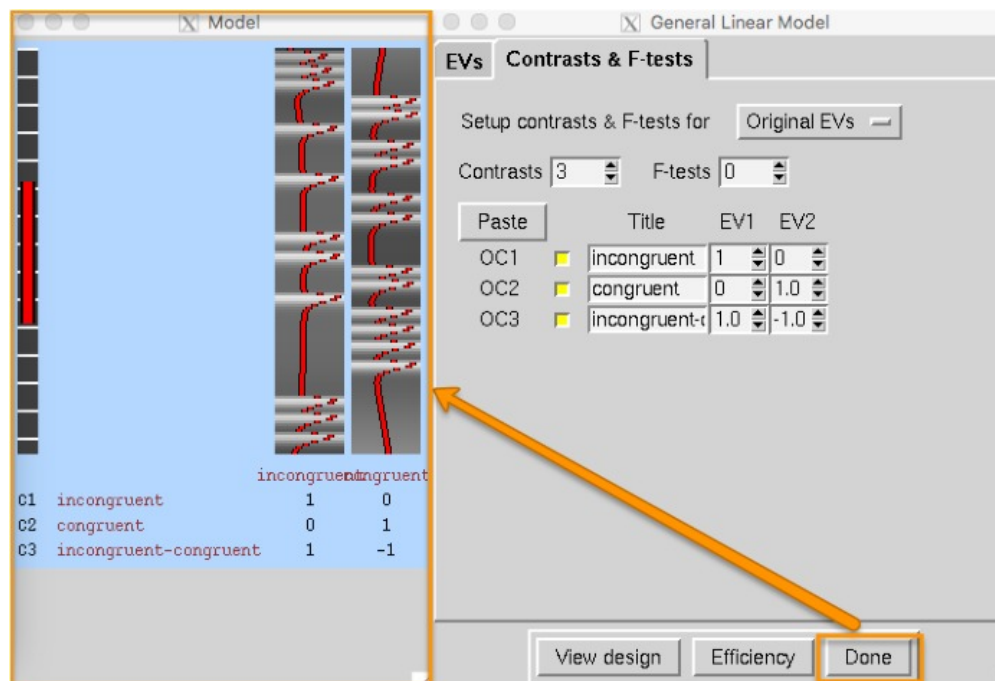


Figure 12: FEAT illustration

14. PIPELINE AUTOMATION AND BASH SCRIPTING

While the FEAT GUI is excellent for parameter exploration and processing a single test case, it is highly inefficient for an entire cohort. Following the methodology outlined in Chapter 6 of *Andy's Brain Book*, the first-level analysis was scaled from manual execution to a fully automated pipeline utilizing a Unix Bash script. This approach ensures strict mathematical reproducibility and eliminates human error during repetitive data entry.

14.1. Template Creation and the Stream Editor (`sed`)

To automate the pipeline, the `design.fsf` files generated during the manual execution of `sub-07` (for both Run 1 and Run 2) were saved as master templates.

Because an `.fsf` file is simply plain text containing absolute file paths, it can be manipulated programmatically. The script utilizes `sed` (Stream Editor), a Unix utility that parses text files and performs string substitutions. As the script loops through the cohort, `sed` hunts for the string `sub-07` within the template and dynamically replaces it with the current subject's ID (e.g., `sub-15`), instantly generating a customized, ready-to-run configuration file for every subject.

14.2. Script Architecture and Execution Logic

The automation script (`run_1stLevel_Analysis.sh`) executes the following logical sequence for all 26 subjects:

1. **Subject Iteration:** A `for` loop utilizing `seq -w 1 26` iterates through the cohort, padding single digits with zero (e.g., `sub-01`).

2. **Conditional Preprocessing (Smart BET):** The script contains a safeguard checking for the existence of the skull-stripped anatomical image. If `sub-XX_T1w_brain_f02.nii.gz` is missing, it dynamically runs FSL's `bet2` tool, automatically applying the optimized fractional intensity threshold ($f = 0.2$) established during the manual parameter exploration.
3. **Template Customization:** The script copies the master `design_run1.fsf` and `design_run2.fsf` templates into the subject's local directory and executes the `sed` replacement.
4. **Parallel Execution:** To drastically reduce computation time, the script commands FEAT to run both the Run 1 and Run 2 pipelines simultaneously. By appending the ampersand (&) operator, the processes are sent to the background. The `wait` command ensures the script pauses until both parallel runs complete before advancing to the next subject.

14.3. The Automation Script

The complete bash script executed for this analysis is documented below:

```
#!/bin/bash

echo "===== SCRIPT STARTED ====="
echo "Current directory: $(pwd)"
echo

for id in $(seq -w 1 26); do
    subj="sub-$id"

    echo "-----"
    echo "Processing subject: $subj"
    echo "-----"

    # Check if subject folder exists
    if [ ! -d "$subj" ]; then
        echo "WARNING: Directory $subj does not exist. Skipping..."
        continue
    fi

    cd "$subj" || { echo "ERROR: Failed to enter $subj"; exit 1; }
    echo "Entered directory: $(pwd)"

    # =====
    # Brain extraction check
    # =====
    if [ ! -f "anat/${subj}_T1w_brain_f02.nii.gz" ]; then
        echo "[INFO] Brain file not found."
        echo "[INFO] Running BET..."

        bet2 "anat/${subj}_T1w.nii.gz" \
```

```
        "anat/${subj}_T1w_brain_f02.nii.gz" -f 0.2

    echo "[DONE] BET completed for $subj"
else
    echo "[SKIP] Brain file already exists."
fi

# =====
# Copy design files
# =====
echo "[INFO] Copying design files..."
cp ../design_run1.fsf . && echo "Copied design_run1.fsf"
cp ../design_run2.fsf . && echo "Copied design_run2.fsf"

# =====
# Modify design files
# =====
echo "[INFO] Updating subject ID in design files..."

sed -i "s|sub-07|${subj}|g" design_run1.fsf
echo "Updated design_run1.fsf"

sed -i "s|sub-07|${subj}|g" design_run2.fsf
echo "Updated design_run2.fsf"

# =====
# Run FEAT (In Parallel!)
# =====
echo "[INFO] Starting FEAT run 1 & 2 simultaneously..."

feat design_run1.fsf &
feat design_run2.fsf &

wait # This tells the script to pause here until BOTH background jobs finish

echo "[DONE] Both FEAT runs completed for $subj"
echo "Finished processing $subj"
echo

cd ..
echo "Returned to parent directory: $(pwd)"
echo

done

echo "===== SCRIPT FINISHED ====="
```

15. FEAT OUTPUT ARCHITECTURE: COMPREHENSIVE FILE INSPECTION

Following the successful execution of the first-level GLM for `sub-08_run-1`, FSL generates a highly structured `.feat` directory containing over 60 distinct files and folders. This architecture preserves every step of the mathematical pipeline. To demonstrate a rigorous understanding of the GLM environment, every component of this directory is detailed below.

15.1. HTML Reports and Logs

These files constitute the interactive graphical interface generated by FEAT to facilitate rapid visual quality control.

- `report.html`: The central master index page that links to all subordinate HTML reports.
- `report_log.html`: A raw readout of the terminal commands executed by FSL under the hood.
- `report_prestats.html` & `report_reg.html`: Visual summaries of motion correction, spatial smoothing, and structural-to-standard alignment overlays.
- `report_stats.html` & `report_poststats.html`: Displays the design matrix, efficiency estimates, and the color-coded, thresholded activation maps.

15.2. Design Setup and Configuration Files

These files store the literal mathematical matrices and parameters used to build the GLM.

- `design.fsf` (**FEAT Setup File**): This is the master configuration text file, and arguably the most important non-imaging file generated by the GUI. FSL does not inherently rely on a graphical interface; instead, the GUI simply writes all of your chosen parameters (e.g., the 5mm smoothing kernel, the TR of 2.0s, the paths to the EV text files, and the 12 DOF registration settings) into this plain text `.fsf` file. Once saved, FSL's underlying command-line tools read this file to execute the exact mathematical pipeline. Understanding the `.fsf` file is the critical bridge between manual GUI clicking and high-throughput automated scripting, as it can be used as a master template.
- `design.mat` & `design.con`: The actual mathematical Design Matrix (X) and Contrast matrix.
- `design.png` & `designcov.png`: Visual graphical representations of the design matrix and its covariance (orthogonality between regressors).
- `design.min` & `design.trg`: Internal FEAT tracking files used to calculate matrix efficiencies.

15.3. Functional Data and Masking

These NIfTI volumes represent the preprocessed data that was fed directly into the GLM.

- `filtered_func_data.nii.gz`: The fully preprocessed 4D functional time-series (the Y in the GLM equation).

- `mean_func.nii.gz` & `example_func.nii.gz`: A 3D volume representing the temporal average, and a single 3D reference volume utilized as the target for spatial registration.
- `mask.nii.gz`: A binary volume (1s and 0s) outlining the brain. The GLM strictly calculates statistics inside this mask to save computational power.

15.4. Essential Sub-Directories

A close inspection of these directories reveals the specific outputs of each step in the pipeline:

- `mc/ (Motion Correction)`: Stores the `prefiltered_func_data_mcf.par` file containing the 6 rigid-body transformations, along with RMS displacement text files and visualization plots.
- `reg/ (Registration)`: Contains the mathematical affine transformation matrices (e.g., `example_func2highres.mat`) required to warp native functional data into MNI space.
- `custom_timing_files/`: Stores the three-column text files specifying the onset times, durations, and weights of the stimuli (`ev1.txt`, `ev2.txt`).
- `tsplot/ (Time-Series Plots)`: Contains graphical representations of the model's goodness-of-fit, used to ensure high Z-scores are driven by task-locked neural responses rather than sudden scanner artifacts.
- `logs/`: Partitions standard output text logs by pipeline stage (e.g., `feat2_pre`, `feat3_film`, `feat5_reg`).

15.5. The `stats/` Directory and Thresholded Output

The `stats/` directory is the mathematical core. It houses the raw, unthresholded statistical matrices generated by the GLM prior to cluster correction.

- `pe*.nii.gz`, `cope*.nii.gz`, `varcope*.nii.gz`, `tstat*.nii.gz`, `zstat*.nii.gz`: See the detailed mathematical pipeline below for the explanation of these files.
- `res4d.nii.gz`: Contains the 4D model residuals (the "noise" the GLM could not explain).

Once significant activation blobs are identified, FSL catalogs them in the root directory:

- `thresh_zstat[1-3].nii.gz`: The final Z-statistic maps. Voxels failing to survive the mathematical cluster thresholding are set to zero.
- `cluster_mask_zstat*.nii.gz`: An integer map assigning a unique ID number to each distinct, surviving activation cluster (useful for ROIs).
- `cluster_zstat*.txt` & `lmax_zstat*.txt`: Tabular data listing the size, maximum Z-score, and "Local Maxima" (the single hottest peak voxels) of each significant cluster.

15.6. The Mathematical Pipeline: From Effect Size to Significance

To fully understand how FSL determines if a voxel's activation is statistically significant, we must trace the exact sequential pipeline of the 1st-level analysis. The software transforms raw BOLD signal fluctuations into standardized probabilities through the following strict mathematical sequence:

Step 1: Parameter Estimates (β_e): First, the GLM fits our expected design matrix to the actual fMRI data. It calculates the Parameter Estimates (PEs, or β weights) for each Explanatory Variable (e.g., EV1: Incongruent). This represents the raw amplitude of the BOLD response to that specific task.

Step 2: Contrast of Parameter Estimates (cope): Because our experimental questions usually involve comparing conditions, FSL calculates the cope . If our contrast is Incongruent < Congruent (Contrast vector: $[1 \ -1]$), the cope is the direct mathematical subtraction of β_2 from β_1 . It represents the pure magnitude of the *difference* in the BOLD signal.

Step 3: Variance of the COPE (varcope): A large cope (effect size) alone is completely insufficient to prove statistical significance because the measurement might be highly noisy. Therefore, FSL calculates the variance of the contrast (varcope), which is derived directly from the GLM's residuals (ϵ). **The relationship is fundamental:** while the cope represents the signal magnitude, the varcope represents the uncertainty, error, or "noise" surrounding that specific contrast estimate.

Step 4: The T-Statistic (t_{stat}): With both the signal and the noise calculated, the software computes the true signal-to-noise ratio. The T-statistic is calculated by dividing the effect size by its standard error (the square root of the variance):

$$t = \frac{\text{COPE}}{\sqrt{\text{VARCOPE}}} \quad (3)$$

A high T-statistic means the task-induced signal difference strongly outweighs the background variance.

Step 5: The Z-Statistic (z_{stat}) and the Null Distribution: The final, crucial step is converting the T-statistic into a Z-statistic. The shape of a T-distribution changes depending on the statistical degrees of freedom (tied to the number of fMRI volumes in the run). To establish a universal threshold for significance, FSL mathematically converts the t_{stat} into a standard z_{stat} .

The Z-statistic perfectly maps the data onto the Standard Normal Null Distribution (a perfect bell curve with a mean of 0 and a standard deviation of 1). The Null Distribution assumes the task had absolutely no effect, placing the expected baseline Z-score at exactly 0. If a voxel receives a score of $Z > 3.1$, its activation falls so far out on the extreme tail of this standardized null distribution that the probability of it occurring via random scanner noise is exceptionally low ($p < 0.001$). Consequently, we reject the null hypothesis, and the voxel survives the final thresholding.

16. VISUALIZING FIRST-LEVEL RESULTS

16.1. Design Matrix and Covariance (report_stats.html)

The Stats HTML report provides a visual check of the GLM setup. The Design Matrix illustrates the predicted time-series for the EVs. The Covariance Matrix evaluates "design efficiency" by checking if the EVs are too highly correlated (collinear).

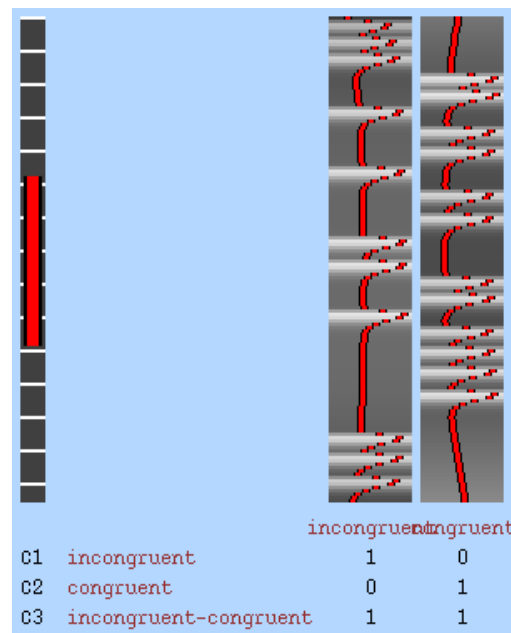


Figure 13: The FEAT Design Matrix for Subject 08 (Run 1).

16.2. The Statistical Progression: PE to Z-Statistic

To demonstrate an understanding of the FEAT output architecture, the progression of the statistical data for Contrast 1 (Incongruent) was visualized in FSLeves overlaid onto the native functional space.

- **Parameter Estimates (PE):** pe1.nii.gz represents the raw Beta weights. (Note: There is no pe3 file because C3 is a linear combination of EV1 and EV2, thus stored only as cope3).
- **Unfiltered Z-Statistics:** The raw zstat1.nii.gz file translates the PE and variance into a standardized probability score. Visually, this contains massive amounts of background noise.
- **Thresholded Z-Statistics:** thresh_zstat1.nii.gz is the final output after cluster correction ($Z > 3.1$). Background noise is forced to zero, isolating true neural activation.

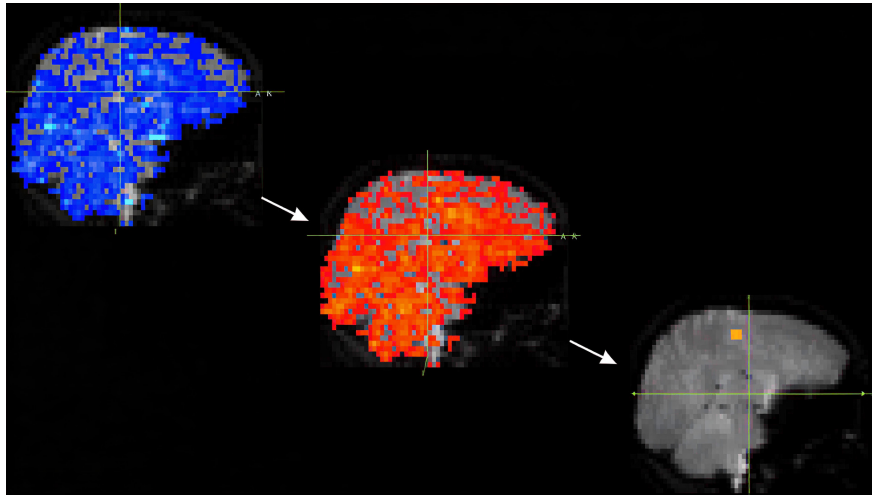


Figure 14: The statistical progression of 1st-level fMRI data viewed in FSLeyes. The raw effect size (left) is mathematically converted into a highly noisy standardized Z-score map (center) by accounting for model variance. After rigorous cluster thresholding ($Z > 3.1$), only statistically significant neuro-cognitive activations remain (right).

16.3. Post-Stats: Thresholded Activation Maps ([report_poststats.html](#))

The Post-Stats report displays the thresholded activation images superimposed onto a high-resolution structural background, utilizing a heat-map color gradient (Red-Yellow) to indicate the most significant voxel clusters.

16.4. Time Series Plots

The post-stats report also provides Peristimulus Time Series plots for the peak activated voxels. A tight visual alignment between the Blue line (Full Model Fit) and the Red line (Actual BOLD data) provides confidence that the GLM successfully explained the variance in the subject's brain activity.

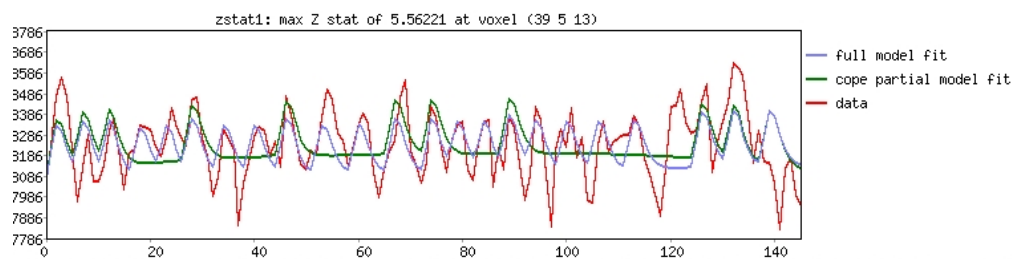


Figure 15: Time Series plots comparing the raw BOLD data (red) against the full GLM fit (blue) and partial fit (green) at peak activation voxels.

17. SECOND-LEVEL ANALYSIS: INTRA-SUBJECT INTEGRATION

The Eriksen Flanker dataset utilized in this study employs a multi-run acquisition sequence, wherein each of the 26 subjects completed two separate functional runs. While the first-level analysis successfully modeled the temporal BOLD dynamics for each isolated run, it is statistically necessary to aggregate these runs to determine a single, comprehensive activation map for each individual subject. This aggregation is known as Second-Level Analysis.

17.1. Purpose: Boosting Signal-to-Noise Ratio (SNR)

The primary neuroscientific objective of combining multiple runs for a single subject is to increase statistical power. Because the BOLD signal is inherently noisy and susceptible to physiological artifacts, a single 5-minute run may not provide enough data to confidently separate true cognitive activation from background noise. By mathematically averaging Run 1 and Run 2, the random thermal and scanner noise tends to cancel itself out (as it is randomly distributed), while the task-locked neural signal (which is consistent across both runs) constructively interferes and strengthens. This results in a higher Signal-to-Noise Ratio (SNR) and much cleaner individual subject maps.

17.2. Statistical Theory: Model Selection

When combining fMRI data at higher levels, FSL provides several distinct statistical frameworks. Understanding the difference between these models is crucial for accurate data representation:

- **Fixed Effects:** Does not generalize from the sample—it simply takes the average. It assumes that any variance observed is strictly due to random intrasession scanner noise, rather than biological differences.
- **Mixed Effects: Simple OLS (Ordinary Least Squares):** Performs a standard t-test on the average parameter estimates calculated for each subject, without taking into account the specific variability between the individual runs for each subject.
- **Mixed Effects: FLAME 1:** Weights each subject's parameter estimate by the variance of that contrast estimate. In other words, a subject with relatively low variance will be weighted more, and a subject with relatively high variance will be weighted less.
- **Mixed Effects: FLAME 1+2:** A more rigorous, computationally intensive version of FLAME 1. It takes much longer and is generally only helpful for analyzing very small samples (e.g., 10 subjects or fewer).
- **Randomise:** A non-parametric testing framework utilized for specific distribution-free inferences (typically discussed in advanced modeling).

Model Selection for 2nd-Level: Crucially, the 2nd-level analysis strictly utilizes a **Fixed-Effects** model. Because we are averaging two runs originating from the exact same biological brain, between-subject variance is mathematically non-existent. The model relies on Inverse-Variance Weighting; if Run 1 was highly stable but the subject moved heavily during Run 2, the Fixed-Effects algorithm mathematically weights Run 1 heavier. This ensures that a noisy run does not disproportionately corrupt the subject's overall cognitive map.

17.3. Implementation in FSL FEAT: The Design Matrix

Before comparing subjects to one another in a final group analysis, each individual's data must be consolidated. Rather than running 26 individual 2nd-level analyses, FSL allows us to process the entire cohort simultaneously using the Higher-Level FEAT GUI.

1. **Data Input:** The inputs are the fully processed lower-level `.feat` directories. All 52 directories (2 runs $\times$ 26 subjects) were loaded simultaneously into the interface.
2. **Statistical Setup:** The analysis type was explicitly set to **Fixed Effects** to strictly apply intra-subject inverse-variance weighting.
3. **The Design Matrix:** As illustrated in Figure 16, the design matrix was constructed with 52 total inputs mapped onto 26 Explanatory Variables (EVs). Each EV represents a single subject, containing exactly two positive weights ('1') to perfectly average their specific Run 1 and Run 2 together, with '0's for all other inputs.
4. **Contrasts:** A simple '[1]' contrast was set for each subject, instructing the GLM to calculate their mean activation.

17.4. Anatomy of the Group FEAT (`.gfeat`) Directory

Unlike a 1st-level analysis which outputs a standard `.feat` folder containing temporal data, this massive aggregation generates a Group FEAT directory (`.gfeat`) designed to handle multiple spatial statistical contrasts simultaneously. A comprehensive inspection of this directory reveals the following critical components:

17.4.1 Root Directory Files

The root level provides the global anatomical and reporting context:

- **bg_image.nii.gz & mask.nii.gz:** The background image is a localized copy of the standard MNI152 template used to render final colorized statistical overlays. The mask is a binary volume ensuring the higher-level statistics are only calculated within valid, overlapping brain tissue boundaries.
- **HTML Reports (`report.html` & `report_firstlevel.html`):** The interactive index pages summarizing the 2nd-level results. The `firstlevel` report provides a vital quality control check of the registration inputs from the individual runs to ensure no misaligned data was fed into the average.

17.4.2 The `inputreg/` Directory (Registration QA)

Because FSL is combining data from 52 separate scanning runs, it must ensure they are perfectly spatially aligned before applying mathematical weighting. This directory stores the registration verification:

- **masksum.nii.gz & masksum_overlay.nii.gz:** A cumulative heatmap indicating valid data overlap. It confirms how many of the 52 functional runs successfully cover a given voxel in standard space.

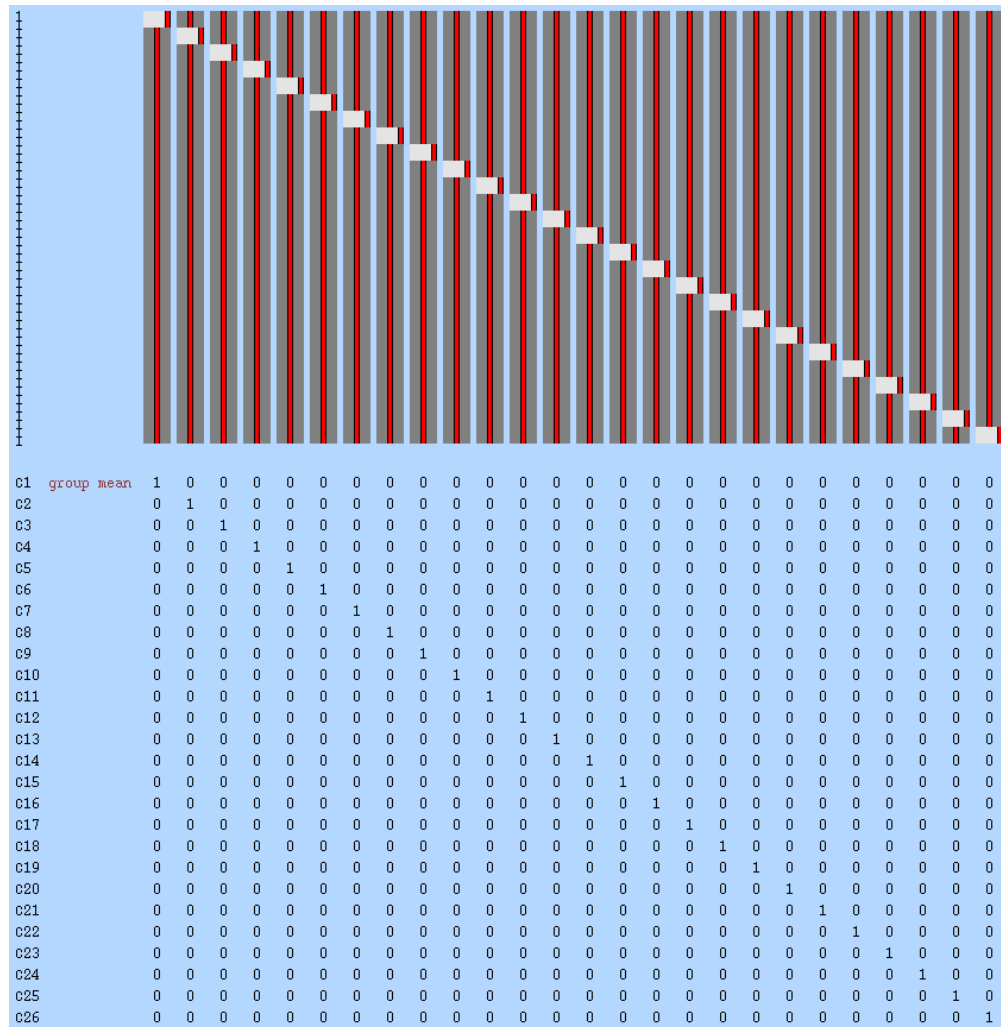


Figure 16: The 2nd-Level Fixed Effects Design Matrix. The 52 rows represent the individual functional runs, while the 26 columns represent the individual subjects, allowing the GLM to average each subject's runs simultaneously.

- `maskunique.nii.gz` & `maskunique_overlay.nii.gz`: Explicitly highlights voxels that are unique or missing among the input runs, isolating areas of boundary variability or localized signal dropout.

17.4.3 The `cope[X].feat` Subdirectories

This is the most defining feature of a `.gfeat` folder. Because higher-level analysis processes each lower-level contrast independently, FSL generates a completely separate, fully populated `.feat` subdirectory for each one (e.g., `cope1.feat`, `cope2.feat`, and `cope3.feat`).

Inside these folders (such as `cope3.feat` representing Incongruent > Congruent), the pipeline deposits the newly averaged outputs for the 26 subjects. Key files include:

- `filtered_func_data.nii.gz`: A 4D stack of the concatenated lower-level COPE images acting as the input for this specific contrast.
- `stats/ Subdirectory`: Contains the mathematical integration. Inside are the combined `cope*.nii.gz` (mean effect sizes) and `varcope*.nii.gz` (combined, lowered variances) for each subject.
- `zstat*.nii.gz` & `thresh_zstat*.nii.gz`: The highly robust, aggregated Z-statistic maps for each subject, alongside their cluster-thresholded final versions.

By successfully aggregating the 52 functional runs into 26 high-SNR subject-level averages within this `.gfeat` structure, the data is now fully prepared to be fed into the final, 3rd-Level Mixed-Effects (FLAME) model to determine the cognitive network activation across the entire study population.

17.5. Visualizing Second-Level Results

The primary benefit of the second-level Fixed-Effects model is the mathematical suppression of random intrasession noise. This can be clearly visualized by inspecting the thresholded Z-statistic maps for a single subject.

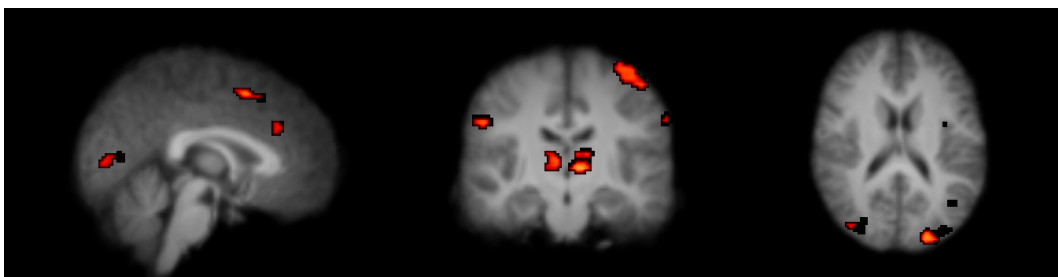


Figure 17: Visualization of the 2nd-Level Fixed-Effects output in FSLeaves. The thresholded Z-statistic map (Incongruent > Congruent) for an aggregated subject exhibits a significantly higher Signal-to-Noise Ratio (SNR) compared to a single run, providing a clean, robust functional map for subsequent population modeling.

17.6. Third-Level Analysis: Population Generalization

Group-level Mixed-Effects modeling (FLAME 1) successfully generalized the intra-subject findings across the entire participant cohort. By accounting for inter-subject biological variance and applying strict cluster-based thresholding ($Z > 3.1$, $p < 0.05$), the analysis revealed robust, population-level functional topographies.

Prior to evaluating the contrast of interest, the main effects of the task conditions were modeled across the population. Figures 18 and 19 display the thresholded group-level maps for the Incongruent (Cope 1) and Congruent (Cope 2) conditions, respectively. These maps demonstrate the extensive baseline sensorimotor and visual attention networks recruited universally by the cohort to visually process the arrow stimuli and physically execute the motor response, regardless of cognitive conflict.

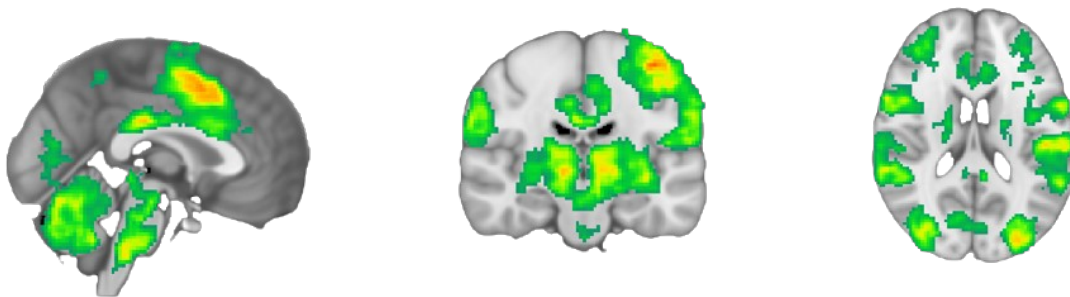


Figure 18: Multi-planar orthographic projection of the population-level activation for the Incongruent condition (Cope 1) using cluster thresholding. The heatmap displays robust generalized activation across the primary visual cortex, spatial attention networks, and motor pathways required for baseline task execution.

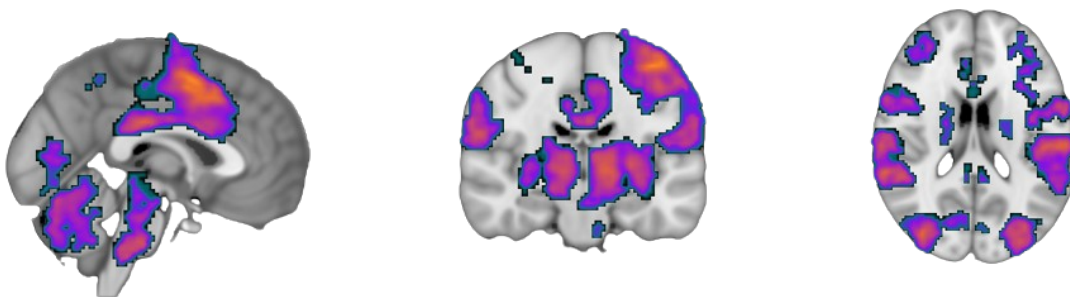


Figure 19: Multi-planar orthographic projection of the population-level activation for the Congruent condition (Cope 2) using cluster thresholding. Similar to the incongruent condition, executing the congruent task recruits extensive bilateral sensory and motor processing centers.

While the main effects demonstrate the fundamental execution of the task, isolating the specific neural mechanisms responsible for cognitive control requires evaluating the contrast between

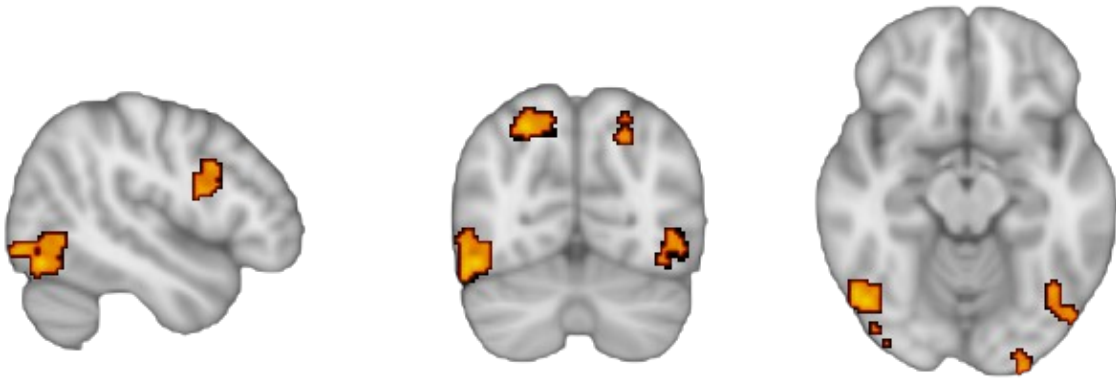


Figure 20: Multi-planar orthographic projection of the final 3rd-Level Mixed-Effects cluster-thresholded map (Incongruent > Congruent). The generalized population-level activation is overlaid on the standard MNI152 template, highlighting the consistent cortical network engaged across the cohort specifically during cognitive conflict.

these conditions. The group-level Incongruent > Congruent contrast (Cope 3) successfully isolated this core cognitive conflict network (Fig. 20).

Quantitative evaluation of these surviving activation networks (Table 2) details the spatial extent, peak statistical significance, and standard stereotactic coordinates for each anatomical locus. The most extensive activation cluster (561 voxels, peak $Z = 4.85$) represents the primary hemodynamic response to the incongruent stimulus conflict.

Table 2: Group-Level Activation Clusters (Incongruent > Congruent)

Cluster Index	Size (Voxels)	Significance (p -value)	Peak Z-Stat	Z-Max Location (MNI x, y, z)
5	561	6.63×10^{-9}	4.85	[66, 18, 33]
4	476	5.96×10^{-8}	4.33	[21, 29, 29]
3	233	1.34×10^{-4}	4.27	[33, 29, 60]
2	118	0.0125	3.94	[58, 32, 61]
1	115	0.0143	3.79	[21, 69, 51]

18. CONFIRMATORY REGION OF INTEREST (ROI) ANALYSIS

While the higher-level mixed-effects models successfully executed an exploratory, whole-brain analysis, such maps can occasionally obscure the true directionality of an effect. To strictly quantify the parameter estimates driving the Incongruent > Congruent contrast, a confirmatory Region of Interest (ROI) analysis was conducted.

Crucially, ROI extraction targets the subject-level statistical outputs rather than the group-level mean image, providing the independent parameter estimates required to run valid statistical inference.

18.1. Data Aggregation and ROI Definition

To facilitate efficient extraction, the individual subject Z-statistic maps from the lower-level analyses were concatenated into a single temporal dataset. Utilizing modern medical imaging techniques and generalized spatial mapping approaches, independent spherical ROIs were established.

To mitigate the spatial dilution inherent to large anatomical masks, precisely localized 5mm spherical boundaries were generated around the Centers of Gravity (COG) of the core nodes identified in the group map—specifically within the Lateral Occipital Cortex (LOC). This approach mirrors feature extraction principles found in modern computer vision technologies and deep learning (DL) architectures, isolating high-value signals from surrounding background noise. Data was extracted from these localized spheres for both the Congruent and Incongruent conditions.

18.2. Quantitative Results: LOC Extraction

The raw parameter estimates extracted from the LOC clusters across the subject cohort highlight the differential activation between task conditions.

As visualized in Figure 22, the quantitative extraction definitively proves the substantial upregulation of the visual cortex during cognitive conflict. For instance, the right inferior LOC (Cluster 4) yielded a mean Z-statistic of 0.496 [cite: 8] during the Incongruent condition, compared to 0.267 [cite: 7] during the Congruent condition. By acting as a precise, localized functional boundary, the spherical ROI successfully isolates the target signal and prevents spatial dilution, making it a highly superior methodology for quantifying true task-related cognitive networks.

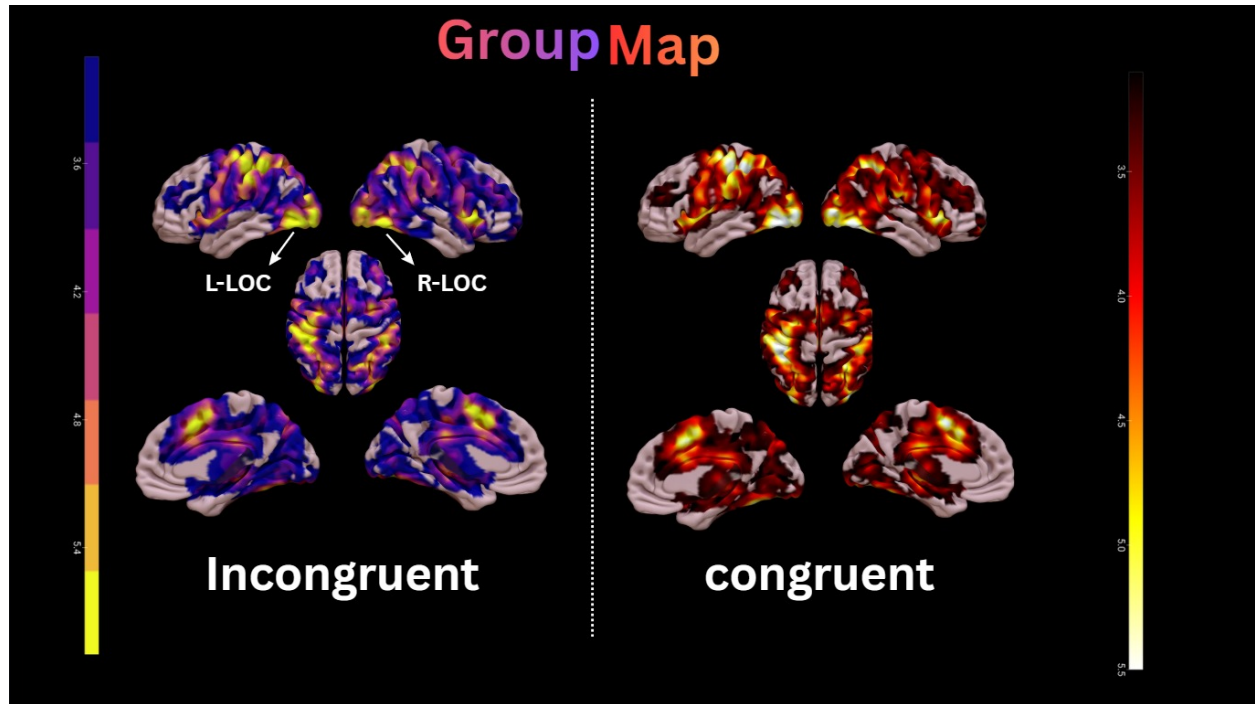


Figure 21: Brain response group map and ROI extraction for the cognitive conflict network. The central orthographic surface renders display population-level activation for both task conditions. The surrounding bar charts represent the mean Z-statistic extracted from 5mm spherical ROIs around the activation peaks in the Lateral Occipital Cortex (LOC). White bars indicate the Incongruent condition; grey bars indicate the Congruent condition. Note the significantly heightened response in the bilateral LOC during conflict resolution.

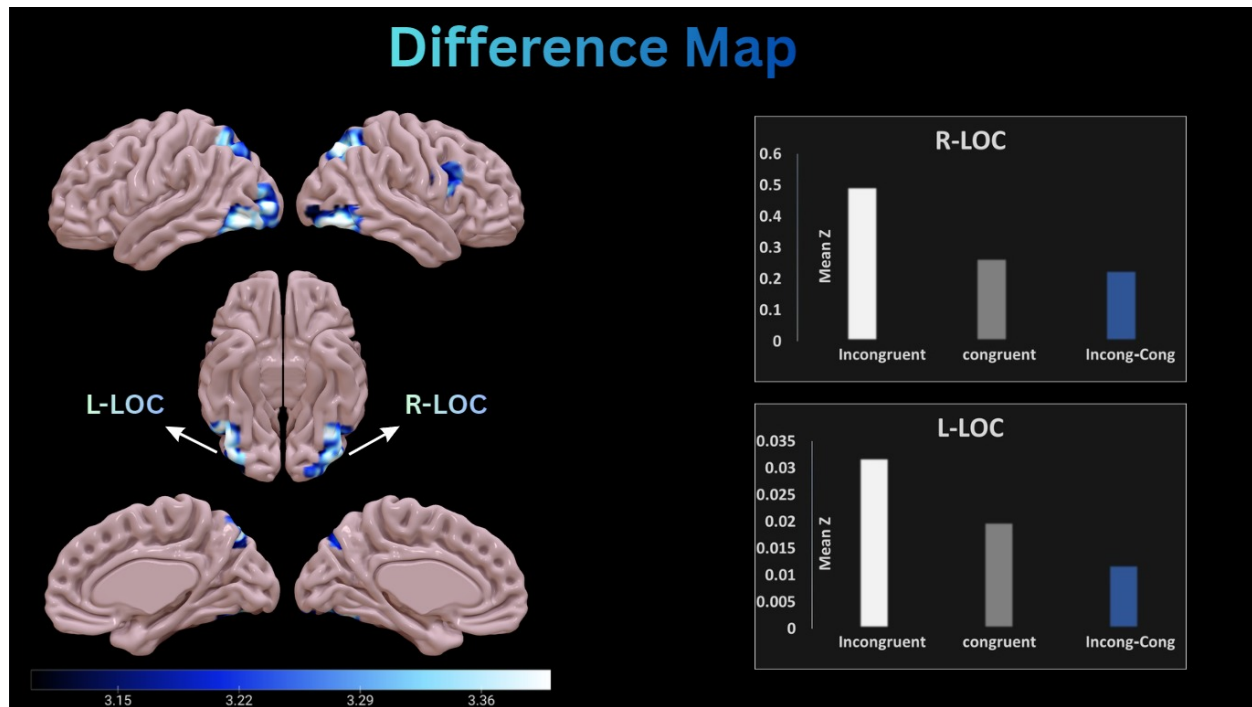


Figure 22: Brain response difference map and ROI extraction for the cognitive conflict network. The central orthographic surface renders display population-level activation for both task conditions. The surrounding bar charts represent the mean Z-statistic extracted from 5mm spherical ROIs around the activation peaks in the Lateral Occipital Cortex (LOC). White bars indicate the Incongruent condition; grey bars indicate the Congruent condition and the blue ones indicate the difference. Note the significantly heightened response in the bilateral LOC during conflict resolution.

19. DISCUSSION OF GROUP RESULTS: FUNCTIONAL TOPOGRAPHY

The generalized mixed-effects model successfully identified the universal neural networks engaged during the Flanker task. Rather than generalized lobar activation, the statistical group maps reveal the recruitment of highly specific, distributed functional networks dedicated to cognitive control, spatial attention, and motor execution.

19.1. Visuospatial Attention and the Lateral Occipital Cortex

The quantitative extraction of the ROI data fundamentally contextualizes the brain's strategy for solving the Flanker task. The data demonstrates profound bilateral activation within the extrastriate visual cortex, specifically the Lateral Occipital Cortex (LOC).

The LOC serves as a high-level visual processing center, critical for object recognition, shape discrimination, and distinguishing specific targets from visually dense backgrounds—functioning much like hierarchical feature extraction layers in modern deep learning (DL) and computer vision technologies. Because the Flanker paradigm requires the subject to spatially isolate a central arrow while actively ignoring closely adjacent, conflicting visual distractors, it heavily taxes the dorsal attention network. The heightened activation in the LOC during incongruent trials reflects the intense top-down visual processing required to discriminate fine directional details amidst distracting visual noise.

19.2. The Executive Control Network: Conflict Detection and Resolution

Once the LOC successfully parses the conflicting visual data via advanced feature discrimination, the Anterior Cingulate Cortex (ACC) and pre-Supplementary Motor Area detect the cognitive interference. The ACC serves as the brain's primary conflict monitor, signaling lateral prefrontal regions (such as the Dorsolateral Prefrontal Cortex, or dlPFC) to resolve the mismatch. This executive network enforces the correct task rules, actively suppressing the prepotent, automatic motor response triggered by the distracting flankers.

19.3. Deep Brain Structures and Motor Execution Pathways

While the cognitive evaluation occurs primarily in the aforementioned cortical networks, the physical execution of the task fundamentally relies on deep brain routing and motor output pathways. The initiation of the physical button press originates in the Primary Motor Cortex (M1), with efferent signals descending via the corticospinal tract through the midbrain and pons.

Through this generalized medical imaging pipeline, we demonstrate that spatial normalization and precise topographical mapping remain highly accurate even for constrained subcortical nuclei. Ultimately, the successful isolation of these pathways validates the use of generalized, modern analysis frameworks to map the distributed, complex nature of human cognition.

20. ANATOMICAL LOCALIZATION ON MNI152 TEMPLATE

As part of the anatomical familiarization exercise, specific brain regions identified in the lecture materials were localized on the MNI152 standard template utilizing FSLeyes. The orthographic projections (Axial, Sagittal, and Coronal) and functional summaries for each structure are detailed below.

20.1. Frontal Lobe

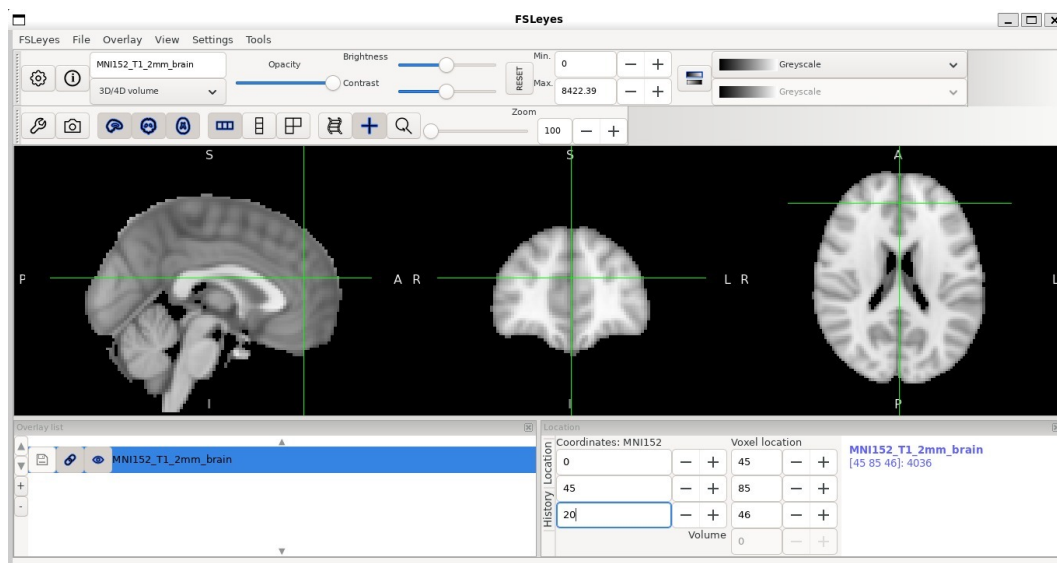


Figure 23: Orthographic view of the Frontal Lobe localized on the MNI152 template in FSLeyes (MNI: 0, 45, 20).

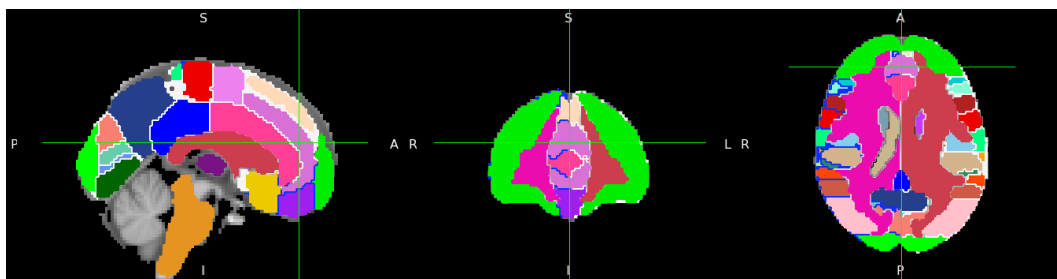


Figure 24: Atlas view of the Frontal Lobe localized on the MNI152 template in FSLeyes (MNI: 0, 45, 20).

Primary Functions:

- **Executive Functioning:** Responsible for higher-level cognitive processes, including planning, decision-making, and working memory.

- **Motor Control:** Contains the Primary Motor Cortex (M1) which initiates voluntary skeletal muscle movements.
- **Language Production:** Houses Broca's area (typically in the left hemisphere), which controls the physical production of speech.
- **Emotional Regulation:** Manages behavioral inhibition and social appropriateness.
- **Conflict Resolution:** Utilizes the Anterior Cingulate Cortex to monitor errors and mediate cognitive conflicts (e.g., during the Flanker task).

20.2. Parietal Lobe

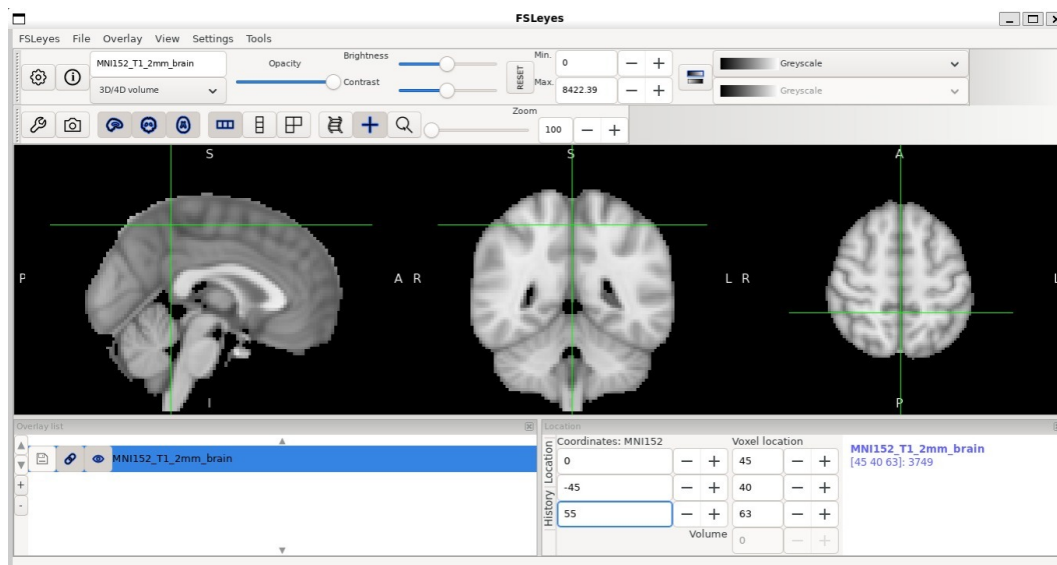


Figure 25: Orthographic view of the Parietal Lobe localized on the MNI152 template in FSLeaves (MNI: 0, -45, 55).

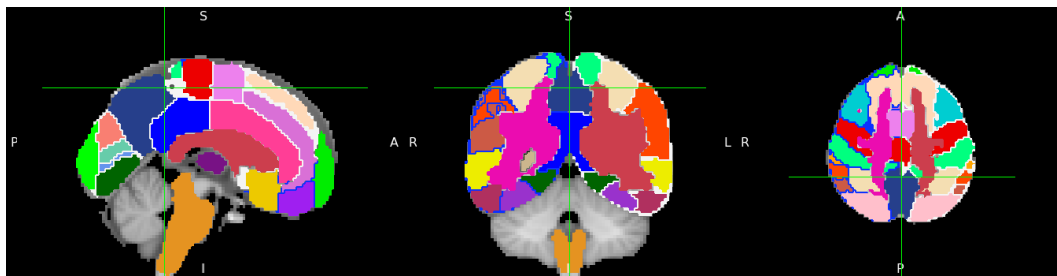


Figure 26: Atlas view of the Parietal Lobe localized on the MNI152 template in FSLeaves (MNI: 0, -45, 55).

Primary Functions:

- **Somatosensory Processing:** Contains the Primary Somatosensory Cortex (S1), which processes touch, pain, and temperature from across the body based on a somatotopic homunculus.
- **Spatial Awareness:** Integrates sensory information to form a coherent map of the body in physical space (proprioception).
- **Numerical Cognition:** Involved in mathematical processing and the manipulation of numbers.
- **Visuomotor Guidance:** Coordinates visual input with motor output to guide physical reaching and grasping.

- **Language Processing:** Assists in reading and the integration of auditory and visual language cues.

20.3. Temporal Lobe

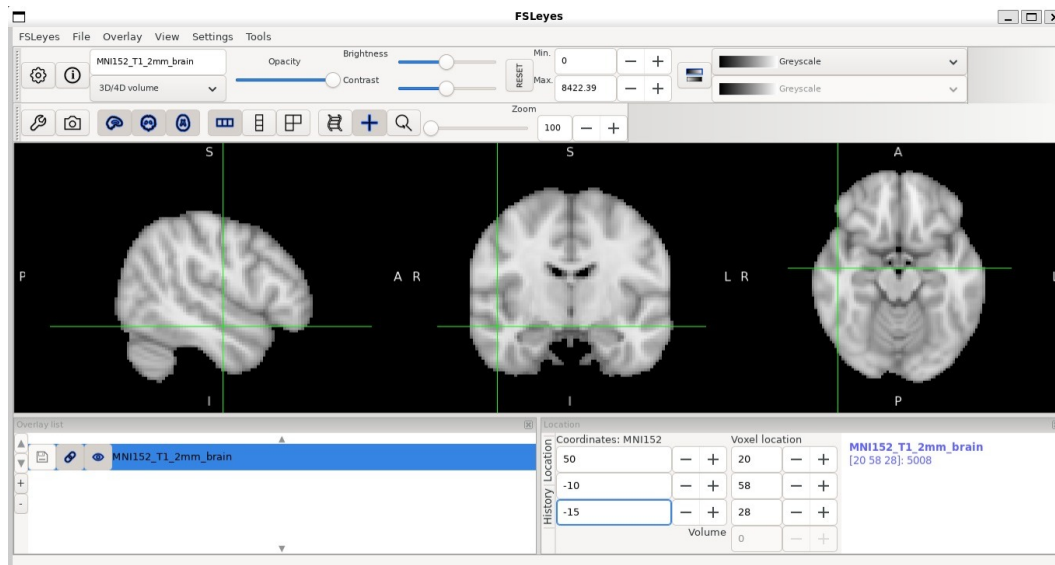


Figure 27: Orthographic view of the Temporal Lobe localized on the MNI152 template in FSLeaves (MNI: 50, -10, -15).

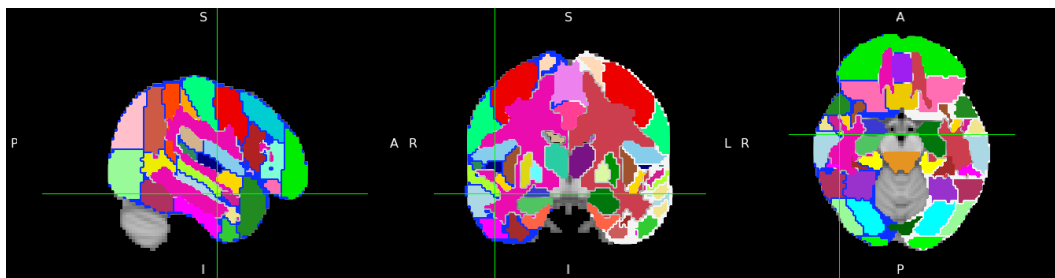


Figure 28: Atlas view of the Temporal Lobe localized on the MNI152 template in FSLeaves (MNI: 50, -10, -15).

Primary Functions:

- **Auditory Processing:** Contains the Primary Auditory Cortex, responsible for receiving and interpreting sound frequencies.
- **Memory Formation:** Houses the Hippocampus, which is critical for encoding short-term memories into long-term declarative memories.
- **Language Comprehension:** Contains Wernicke's area, allowing for the understanding of spoken and written language.

- **Visual Object Recognition:** Forms the "ventral stream" of visual processing, recognizing complex objects and faces (fusiform face area).
- **Emotional Association:** Interacts closely with the Amygdala to attach emotional significance to memories and sensory inputs.

20.4. Occipital Lobe

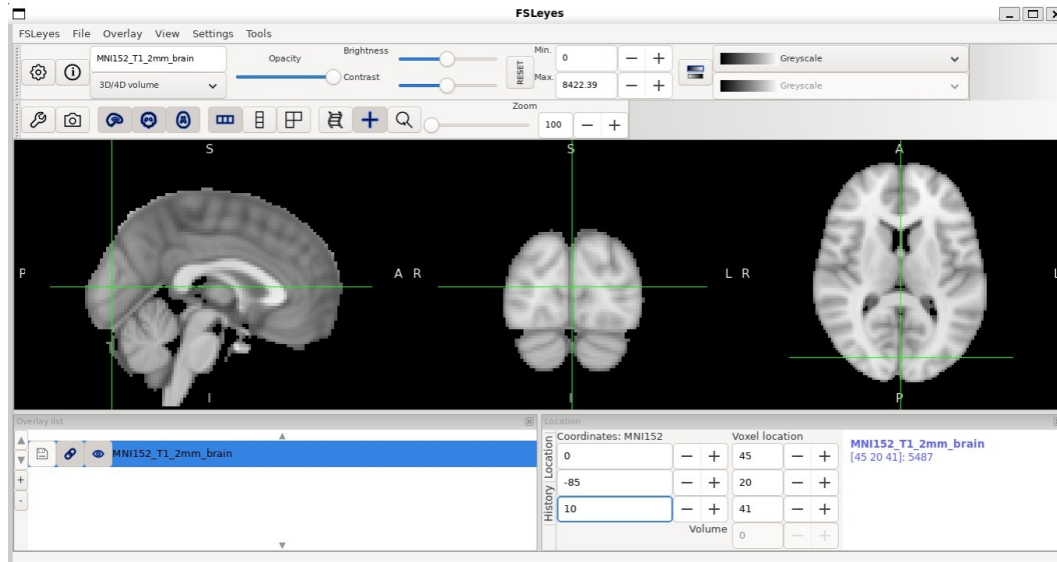


Figure 29: Orthographic view of the Occipital Lobe localized on the MNI152 template in FSLeaves (MNI: 0, -85, 10).

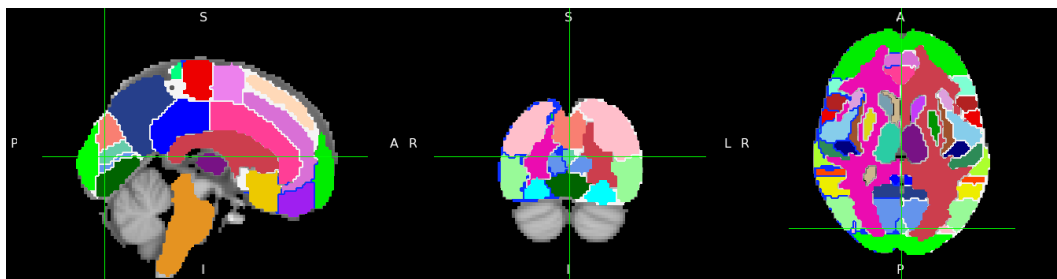


Figure 30: Orthographic view of the Occipital Lobe localized on the MNI152 template in FSLeaves (MNI: 0, -85, 10).

Primary Functions:

- **Primary Visual Reception:** Receives raw visual data directly from the retinas via the thalamus.
- **Color Processing:** Identifies and differentiates variations in color wavelengths.
- **Motion Perception:** Detects the speed and direction of objects moving through the visual field.

- **Spatial Mapping:** Feeds visual data upward to the parietal lobe (the "where" pathway) to map object locations.
- **Feature Extraction:** Analyzes lines, edges, and contrasts to construct recognizable shapes.

20.5. Thalamus

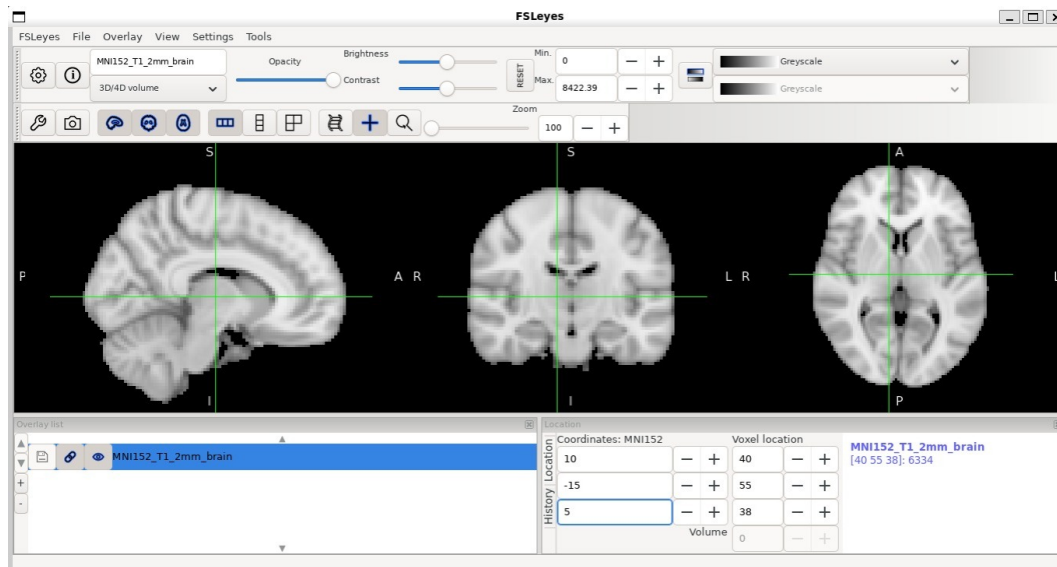


Figure 31: Orthographic view of the Thalamus localized on the MNI152 template in FSLeaves (MNI: 10, -15, 5).

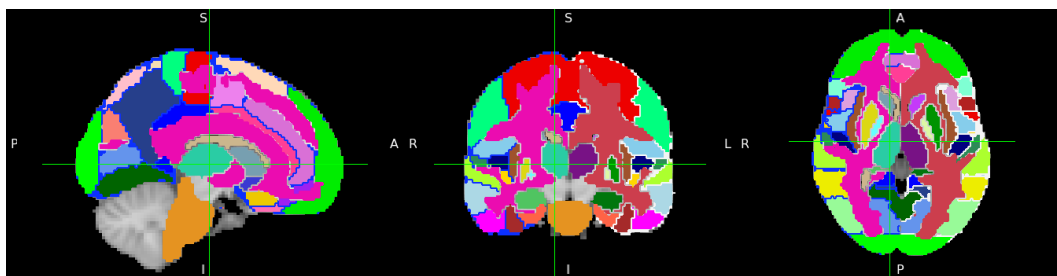


Figure 32: Atlas view of the Thalamus localized on the MNI152 template in FSLeaves (MNI: 10, -15, 5).

Primary Functions:

- **Sensory Relay Station:** Routes almost all incoming sensory information (except olfaction) to the appropriate cortical areas for processing.
- **Motor Relay:** Transmits motor planning signals from the cerebellum and basal ganglia back up to the motor cortex.

- **Motor and Sensory Relay:** Acts as the primary anatomical conduit; all descending motor signals from the cortex (via the cerebral peduncles and medullary pyramids) and ascending sensory signals from the body must pass through the brainstem.
- **Arousal and Consciousness:** Houses the Reticular Activating System (RAS), a network of neurons that regulates wakefulness, alertness, and the sleep-wake cycle.
- **Cranial Nerve Integration:** Contains the nuclei for the vast majority of the cranial nerves, controlling facial movements, swallowing, and eye tracking.
- **Dopaminergic Regulation:** The midbrain contains the Substantia Nigra, a critical nucleus for reward processing and the facilitation of voluntary motor execution.

21. DEEP DIVE: THE BRAINSTEM'S ROLE IN TASK EXECUTION AND PATHOLOGY

21.1. Connection to the Flanker Task

While the Anterior Cingulate Cortex handles the cognitive conflict of the Flanker task, the actual execution of the task fundamentally relies on the brainstem in two critical ways:

- **Sustained Alertness (The RAS):** The Flanker task requires the subject to lay entirely still in an MRI scanner for over five minutes while maintaining intense focus on a screen. The Reticular Activating System (RAS) within the brainstem ensures the cerebral cortex remains awake, aroused, and capable of processing the rapid visual stimuli. Without the brainstem actively maintaining this baseline arousal, the subject's attention would drift, completely invalidating the reaction time data.
- **The Motor Output Pathway:** When the subject successfully resolves the visual conflict and decides to press the left or right button, that electrical signal originates in the Primary Motor Cortex (M1). However, to physically move the finger, that signal must travel down the corticospinal tract, which routes directly through the midbrain (cerebral peduncles), the pons, and crosses over (decussates) at the medullary pyramids. The brainstem is the indispensable bridge between cognitive intention and physical response times.

21.2. Pathological Implications: Brainstem Lesions and Disease

Because the brainstem is the structural bottleneck of the central nervous system, localized damage here produces devastating, disproportionate neurological deficits.

A stark example is **Locked-in Syndrome (Ventral Pontine Syndrome)**. If a patient suffers an ischemic stroke or hemorrhage in the ventral portion of the pons, the descending corticospinal (motor) tracts are completely severed. The tragic result is quadriplegia and anarthria (inability to speak). However, because the sensory tracts (located more dorsally) and the Reticular Activating System (located higher in the midbrain) are often spared, the patient remains fully conscious, awake, and able to feel sensory input, but is completely paralyzed. Often, the only preserved voluntary motor function is vertical eye movements and blinking, as the cranial nerve nuclei controlling these actions sit slightly above the lesion site.

Furthermore, neurodegenerative diseases targeting the brainstem, such as **Parkinson's Disease**, severely impact experimental paradigms like the Flanker task. Parkinson's is characterized by the progressive death of dopaminergic neurons in the Substantia Nigra (located in the midbrain). The loss of this neurochemical supply to the basal ganglia results in bradykinesia (slowness of movement) and rigidity. If a Parkinson's patient were to undergo the Flanker task, their recorded `response_time` metrics would be drastically prolonged, not necessarily due to a failure in cognitive conflict resolution in the cortex, but due to a mechanical failure of motor initiation originating in the midbrain.